

The Winner’s Curse in Data-Driven Decision Making: Evidence and Solutions

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Data-driven decision making involves estimating the value of each potential option and selecting the one with the highest estimated efficacy. This approach underpins a wide array of modern marketing, operations, and AI applications, including A/B testing, advertising and bidding, pricing, and personalized targeting. However, several papers have shown that the estimated effectiveness of the chosen options will be systematically over-optimistic (Smith and Winkler 2006, Efron 2011, Andrews et al. 2024), even when the estimated outcomes are themselves unbiased and efficient. Using simulations calibrated to realistic parameter values from recent marketing studies, we first demonstrate that the magnitude of the winner’s curse is often high in relevant marketing contexts, and that the severity of the winner’s curse depends on the true performance difference between options relative to the level of noise in the data, the number of alternatives under consideration, and the number of observations per tested condition. We further show that using machine learning methods to evaluate what treatment to give to individual consumers can lead to extremely high levels of winner’s curse, especially if the machine learning functional form is very flexible. We propose a correction method based on a non-continuous bootstrap, and benchmark our method against several existing proposed solutions across many common marketing scenarios. We demonstrate that our bootstrap approach generally performs well, and usually outperforms the solutions that have been previously proposed in the literature.

Key words: Winner’s Curse, Data-Driven Decision Making, A/B Testing, Personalized Targeting, Machine Learning

1. Introduction

Data-driven decision making, where a researcher first estimates the expected effectiveness of a set of possible decisions, and then chooses the option that is estimated to be the most effective, is observed across many settings in marketing and AI-driven decision systems. Examples of this *inference-then-optimization* framework includes running A/B tests to test different customer stimuli, and using machine learning to implement personalized advertising and promotional targeting strategies.

In practice, it is crucial to identify not only the optimal actions themselves, but to obtain accurate estimates of the effectiveness of these optimal decisions, as well. For example, online personalized

advertising platforms, such as Meta and Google, first estimate the value of each advertisement and then adjust the advertiser’s effective bid based on the effectiveness of the selected winning ad. Thus, the estimated value of the winning advertisement not only determines which ad is chosen, but also influences the payment made by the winning advertiser (Fan et al. 2022, Tadelis et al. 2023). In other contexts, precisely measuring the effects of chosen treatments can be essential for purposes of internal decision making, resource planning, and even public reporting of financial disclosures related to anticipated outcomes. Similarly, decisions to engage in personalized advertising and promotions can require large fixed cost investments when partnering with advertising platforms. Thus, deciding whether to engage in personalized advertising at all requires obtaining a reasonable estimate of the lift the firm obtains from personalized advertising. Even in a simpler setting, where an advertiser uses an A/B test to both choose which creative is best as well as whether the best advertisement has a high enough return to justify spending on advertising on a particular platform at all, requires a reliable estimate of the return on investment (ROI) for the best-performing stimuli.

The intuitive way to evaluate the likely payoff of the chosen option would be to use the estimated efficacy of that option, assuming that it is unbiased and efficient. However, several papers from statistics, biostatistics, microeconomics and computer science, including Smith and Winkler (2006), Efron (2011), Xu et al. (2011), Lee and Shen (2018), Kuchibhotla et al. (2022), Andrews et al. (2024), show that these estimates will be over-optimistic. This overestimation, known as the winner’s curse,¹ arises from two fundamental features of the framework. First, estimation errors are inevitable in the inference stage, as a perfect estimation of each option’s value is infeasible with finite data samples. Second, the optimization stage involves selecting the option with the highest estimated value, based on these imperfect estimates. The action whose effectiveness is overestimated is more likely to have a higher estimated outcome, so the act of choosing actions based on the estimated impacts introduces estimation bias. Note that because this logic relies on the presence of estimation errors, the winner’s curse is a finite-sample problem. The size of this winner’s curse is generally largest when the effectiveness of each of the top options is similar to each other, when there are relatively few observations in the data (although we will show that this can include cases with tens or even sometimes hundreds of thousands of observations per option), or when there are a lot of available actions. Ultimately, the overestimate of the effectiveness of the chosen action can be much larger than the difference in the effectiveness of the different actions.

The winner’s curse can be especially severe in modern AI-driven decision systems. A defining characteristic of these systems is the massive scale of candidate actions. For example, the recommendation engine from large platforms, such as YouTube, Instagram, and TikTok, selects the

¹ Smith and Winkler (2006) call this the optimizer’s curse.

top-scoring contents from a pool of billions. As we show, making choices from such an enormous action space greatly amplifies overestimation. A second characteristic is the use of highly flexible learning models. When firms leverage large datasets to personalize decisions, they often deploy highly flexible models such as deep causal forests or deep neural networks. While such flexible models reduce approximation bias in capturing complex heterogeneity, our paper finds that they also magnify the winner’s curse due to higher estimation variance.

Other authors have proposed solutions to this winner’s curse. Common solutions include using empirical Bayesian modeling (Efron 2011), selective inference (also known as post-selection inference) (Andrews et al. 2024), and sample splitting (Cox 1975). However, we demonstrate that each of these proposed solutions has issues that are problematic in common marketing contexts.

To overcome these limitations, we develop a bootstrap-based method to correct for the winner’s curse. Specifically, we generate bootstrap samples by resampling with replacement from the original dataset, reconstructing the optimal decisions within each bootstrap iteration, and then evaluating those optimal decisions on both the bootstrap samples and the original dataset. This procedure approximates the winner’s curse by comparing the estimated decision values based on the bootstrap samples to those based on the original data, thereby allowing us to estimate the adjustment needed to mitigate the bias. One complication is that the optimization procedure used to select the optimal decision is typically non-smooth with respect to the underlying data-generating process, which may cause the standard bootstrap to converge more slowly and introduce bias in finite samples (Bickel and Freedman 1981, Fang and Santos 2019). To address this issue, we utilize a subsampling bootstrap technique as a supplement to the standard approach called the m -out-of- n bootstrap (Bickel et al. 1997, Chakraborty et al. 2013), which helps smooth the estimator and can ultimately outperform the standard bootstrap.

Thus, our paper has two goals and in turn contributions. First, we show that the impact of the winner’s curse can be very large in two key scenarios—A/B testing and personalized targeting—under realistic parameters. Second, we benchmarked all commonly proposed solutions in the literature with our proposed simple bootstrap-based and m -out-of- n bootstrap-based solutions, and show that these two methods generally perform well in correcting the winner’s curse across all of the scenarios considered in this paper. We observe that our methods are flexible and do not require strict functional form assumptions to hold in order for the correction techniques to be valid. In general, our m -out-of- n bootstrap approach dominates the performance of empirical Bayes and selective inference approaches. We further demonstrate that while sample splitting provides a better average correction for the winner’s curse than the m -out-of- n bootstrap, (1) using sample splitting methods requires making suboptimal choices that decrease firm profits (so it is expensive to apply in a way that the other fixes are not), and (2) the confidence intervals from sample splitting tend

to be very large, often encompassing the confidence intervals of the bootstrap approach, meaning that the accuracy of the sample splitting fix can be low even if there is not a bias. Ultimately, our findings offer actionable guidance for firms on how to diagnose and correct overestimation in data-driven decision systems, especially in AI-driven environments, where both the number of candidate actions and the flexibility of underlying models can significantly exacerbate the winner’s curse.

The paper is organized as follows. We first present the theory of the winner’s curse in Section 2. We then introduce various proposed remedies from the literature, in addition to our proposed bootstrap-based estimator, in Section 3. Sections 4 and 5 present extensive simulations that demonstrate the presence of substantial winner’s curse and compare the performance of different estimators in correcting it across the two settings: A/B testing and personalized targeting. These two sections discuss applications with continuous outcomes such as customer spending, while Section 6 demonstrates the robustness of the results to applications with discrete outcomes, such as when customers make discrete purchase decisions. Section 7 then concludes the paper.

2. The Theory of Winner’s Curse

The winner’s curse has been studied across various fields of literature. The concept was first introduced by three petroleum engineers who found that the winners of oil-field auctions tended to overbid for these rights (Capen et al. 1971). The reason for this overbidding was that each oil company would send out its geologists to estimate how much oil was in the tract that was being bid on. However, estimating oil reserves is an uncertain science, so some estimates would be too high while others would be too low. As a result, the winning company tended to be the firm that had the highest estimate since that company would be the most aggressive bidder. The solution to the problem was that all firms should shade their bids according to how much they would overestimate the amount of oil in each tract, conditional on their estimate being the highest. Economists subsequently formalized the winner’s curse within the framework of auction theory, extending its application beyond auctions to numerous other economic domains, such as labor economics (Bijkerk et al. 2021).

Beyond economics, the winner’s curse has also been discussed in biostatistics and stochastic systems. In Genome-Wide Association Studies, researchers test millions of genetic variants to identify a single-nucleotide polymorphism associated with certain diseases. However, the estimated correlation is generally found to be smaller in subsequent replication studies (Zöllner and Pritchard 2007, Zhong and Prentice 2008, Xu et al. 2011, Forde et al. 2023). In stochastic systems, Mannor et al. (2007) documented a positive bias when estimating the value function in an infinite-horizon discounted-reward Markov decision process. Furthermore, practitioners have also found such over-optimism in online experiments from different platforms. For example, studies from Airbnb (Lee and

Shen 2018) and Amazon (Kessler 2024) provide evidence that actions selected from statistically-significant A/B tests have estimated total effects that are higher than what the firms got when the tests were launched to the entire user base.

We now formalize the inference-then-optimization framework and, in turn, the winner’s curse phenomenon in a marketing context. Let $\mathcal{A} = \{a_1, \dots, a_J\}$ denote a finite set of J possible marketing actions, and let $j \in \{1, \dots, J\}$ index each action. Define τ_j as the expected value of taking action $a_j \in \mathcal{A}$:

$$\tau_j = \mathbb{E}[Y_i(a_j, \epsilon_i(a_j))] \in \mathbb{R},$$

where $Y_i(a_j, \epsilon_i(a_j))$ represents the realized outcome of taking action a_j for customer i , and $\epsilon_i(a_j)$ captures idiosyncratic shocks that are assumed to be i.i.d. across individuals. We impose no structural assumptions on either $Y_i(\cdot)$ or $\epsilon_i(a_j)$, allowing the data-generating process (DGP) to encompass a wide range of marketing applications.

For example, in a traditional A/B testing setting, marketers may wish to compare two ad creatives to determine which yields higher engagement. In this case, the decision space is $\mathcal{A} = \{a_1, a_2\}$, and each creative has an expected number of clicks denoted by τ_1 and τ_2 , respectively. Similarly, the framework can be extended to a targeting scenario in which marketers choose among multiple price levels for different customer segments. Suppose there are three price levels for each of two customer segments; then the decision space becomes $\mathcal{A} = \{a_1, a_2, \dots, a_9\}$, where each action represents one of the $3^2 = 9$ possible targeting strategies, and each strategy has a corresponding expected revenue $\tau_1, \dots, \tau_9$.

The true expected performance value of each action is unknown to the marketer *a priori*; therefore, it must be estimated from a collection of historical customer responses to each action, obtained either from an observational dataset or a randomized controlled trial. We assume that there are N customer responses for each action, and denote this historical collection as $\mathcal{D}$. Based on this dataset, we assume that the marketer can construct consistent and unbiased estimators $\hat{\tau}_j$ for each option $a_j \in \mathcal{A}$, such that $\mathbb{E}_{\mathcal{D}}[\hat{\tau}_j] = \tau_j$ for all $j \in \{1, \dots, J\}$.

The inference-then-optimization framework assumes that the marketer selects the action with the highest estimated expected reward. We denote this action as the empirical winner:

$$\hat{j}^* \in \arg \max_{j \in \{1, \dots, J\}} \hat{\tau}_j. \quad (1)$$

After selecting the empirical winner, the marketer is interested in evaluating the achieved performance under the inference-then-optimization framework. A common practice is to report the

² When there are ties, we select the action with the smallest index as the winner, without loss of generality.

estimator $\hat{\tau}_{j^*}$ of the empirical winner, since it is an unbiased estimator of τ_{j^*} before selection. However, as shown theoretically in Lemma 1, this reported value—despite being an unbiased estimator—*overestimates* the expected reward obtained from following the inference-then-optimization framework, a phenomenon known as the *winner’s curse*.³

LEMMA 1 (Winner’s Curse). *Given unbiased estimators $\hat{\tau}_j$ for the expected reward (τ_j) of each option $a_j \in \mathcal{A}$, the estimator $\hat{\tau}_{j^*}$ of the empirical winner $j^* = \arg \max_{j \in \{1, \dots, J\}} \hat{\tau}_j$ overestimates its true performance:*

$$WC := \mathbb{E}_{\mathcal{D}}[\hat{\tau}_{j^*} - \tau_{j^*}] \geq 0, \quad (2)$$

This lemma is similar to Smith and Winkler (2006) and intuitive because, when reporting the empirical winner’s expected reward, we are effectively conditioning on the event that this particular estimate is already greater than all others in the action space. Such a selection introduces bias into the original estimator. In other words, although $\mathbb{E}_{\mathcal{D}}[\hat{\tau}_j] = \tau_j$ for all $j \in \{1, \dots, J\}$, it follows that $\mathbb{E}_{\mathcal{D}}[\hat{\tau}_{j^*}] = \mathbb{E}_{\mathcal{D}}[\hat{\tau}_j \mid \hat{\tau}_j \geq \hat{\tau}_k, \forall k \neq j] > \tau_j$. Another way to interpret this result is that, since we cannot disentangle whether the empirical winner truly exhibits superior performance or merely received positive idiosyncratic shocks in the estimation process, the expected reward of the empirical winner may be overestimated simply because it was “lucky” to have received favorable idiosyncratic shocks. Note that the winner’s curse implies that the estimated reward of the empirical winner under the inference-then-optimization framework is a biased estimator of its true intrinsic reward. This does not guarantee that the empirical winner is the true optimal action under the data generating process; in fact, the marketer may select a suboptimal action and still overestimate its expected value in this way.

Given that we have established the general existence of the winner’s curse under the inference-then-optimization framework, we next examine the factors that determine its magnitude. To this end, we consider a concrete setting inspired by a standard A/B testing problem. Specifically, suppose there are two actions $\mathcal{A} = \{a_1, a_2\}$ available to the marketer, and that each action yields a realized reward for individual i given by a linear combination of the true deterministic reward and a Gaussian random shock with zero mean and standard deviation σ , i.e., $Y_i(a_j, \epsilon_i) = \tau_j + \epsilon_i$, where $\epsilon_i \sim \mathcal{N}(0, \sigma^2)$. Without loss of generality, we assume $\tau_2 \geq \tau_1$ and define the reward gap as $\Delta\tau = \tau_2 - \tau_1$. We further assume that the marketer conducts an A/B test in which each action is randomly assigned to N customers, and then follows the inference-then-optimization framework in reporting the empirical winner as well as its expected reward. In this setting, we derive a closed-form expression for the winner’s curse:

³ Smith and Winkler (2006) shows a related result in a different setting that relies on stronger distributional assumptions. We extend their result to a setting without any assumptions on the data generating process, using a more general proof based on Jensen’s inequality.

LEMMA 2 (Winner’s Curse in A/B Testing). *Consider two options $\mathcal{A} = \{a_1, a_2\}$ and the reward of taking each action j on person i is $Y_i(a_j, \epsilon_i) = \tau_j + \epsilon_i$, where $\epsilon_i \sim \mathcal{N}(0, \sigma^2)$. Suppose that we observe N customers randomly assigned to each action and follow the inference-then-optimization framework in Eq. 1. Then the winner’s curse defined in Eq. 2 is:*

$$WC = \sigma \sqrt{\frac{2}{N}} \varphi \left(\frac{\Delta\tau}{\sigma} \sqrt{\frac{N}{2}} \right), \quad (3)$$

where φ is the standard normal density function.

The closed form expression in Eq. 3 highlights that key determinants of the severity of the winner’s curse are the sample size (N), the true reward gap ($\Delta\tau$) and the outcome noise level (σ). As $\Delta\tau$ decreases, the two actions become increasingly indistinguishable, and the marketer’s selection will not be driven by the intrinsic superiority of the actions, but rather by random favorable estimation error. Likewise, for large σ , idiosyncratic shocks have a greater impact on selection, increasing the risk that the empirical winner benefited from noise.

These two forces counterbalance each other. To formally express their joint impact, consider the *relative* winner’s curse, defined as $\frac{WC}{\Delta\tau}$, which measures the degree of overestimation per unit of true lift. Specifically, we have

$$\frac{WC}{\Delta\tau} = \frac{\sigma}{\Delta\tau} \sqrt{\frac{2}{N}} \varphi \left(\frac{\Delta\tau}{\sigma} \sqrt{\frac{N}{2}} \right) = f \left(\frac{\Delta\tau}{\sigma}, N \right),$$

where $f(\cdot)$ is a deterministic function. This expression shows that the joint effect of the reward gap and uncertainty on the relative winner’s curse can be succinctly summarized by a single variable, $\frac{\Delta\tau}{\sigma}$, which we refer to as the **signal-to-noise ratio**. The relative winner’s curse increases as $\frac{\Delta\tau}{\sigma}$ decreases. Owing to this clean and interpretable relationship, *we use the signal-to-noise ratio as the primary tuning parameter and report the relative winner’s curse in our simulation experiments in Sections 4 through 6.*

Beyond the signal-to-noise ratio, the number of candidate actions also plays a critical role. Intuitively, as the marketer considers more alternatives, the probability that at least one action is substantially overestimated increases, thereby magnifying the winner’s curse. To illustrate this point theoretically, let us consider a simple example where all candidate actions have the same expected performance. In this case, we can demonstrate that the winner’s curse increases with the number of candidates, which is formally summarized in the following lemma.

LEMMA 3 (Winner’s Curse with Multiple Action Candidates). *Consider J actions $\mathcal{A} = \{a_1, \dots, a_J\}$ with identical true expected rewards $\tau_1 = \dots = \tau_J = \tau$, such that the outcome for each customer is $Y_i(a_j, \epsilon_i) = \tau + \epsilon_i$, where $\epsilon_i \sim \mathcal{N}(0, \sigma^2)$. Suppose that we observe N customers randomly*

assigned to each action and follow the inference-then-optimization framework in Eq. 1. Then, the total winner’s curse defined in Eq. 2 satisfies:

$$WC = \frac{\sigma}{\sqrt{N}} \left[\sqrt{2 \log J} + o(1) \right].$$

This is simple extension to Lemma 2. The main intuition behind this result is that when all candidate options have the same expected value, selecting the winner is equivalent to selecting the maximum among the estimated expectations of the candidate actions. Then, according to extreme-value asymptotics for Gaussian maxima (Leadbetter et al. 2012), this maximum scales with $\sqrt{\log J}$. This relationship highlights a fundamental scaling property: as the number of candidate actions J increases, the likelihood that random noise alone produces a spurious “winner” grows at a rate of $\sqrt{\log J}$, amplifying the overall severity of the winner’s curse. Although this specific derivation assumes identical true values across candidates, the intuition that the magnitude of the winner’s curse increases with the number of options remains valid more generally.

3. Correcting the Winner’s Curse

Given that we have proved the presence of a winner’s curse, we now consider potential ways to correct for it. We first introduce our bootstrap-based correction method for removing the winner’s curse, which we suggest is a pliable and reasonably robust approach to use across different settings. We then summarize three alternative remedies from the literature: sample splitting, bayesian methods, and selective inference.

3.1. Bootstrap-Based Correction

In auction theory, the solution to the winner’s curse for the bidders is to shade their bid by a small amount. We follow an analogous idea: we debias the estimator by subtracting the bias:

$$\hat{\tau}_{j^*} - WC. \tag{4}$$

Thus, if we can get a good estimate of the winner’s curse, then we can get an estimate that is closer to that of the true effect.

We propose using a bootstrap approach (Efron 1979) to estimate the unknown winner’s curse bias, WC . The problem with estimating the winner’s curse is that we do not know the true parameters for our optimization. However, we can approximate the winner’s curse by taking our estimated effect as a ground truth scenario, simulating a new dataset from these parameters, and then calculating the size of the winner’s curse in such a dataset.

Formally, let $b \in \{1, \dots, B\}$ denote the index of B bootstrap samples, $\hat{\tau}_{b,j}$ denote the estimated reward for the j^{th} action, $a_j \in \mathcal{A}$, in the b^{th} bootstrap sample, and $\hat{j}_b^* \in \arg \max_{j \in \{1, \dots, J\}} \hat{\tau}_{b,j}$ denote

Algorithm 1 Standard Bootstrap Correction

Require: Data $\mathcal{D}$; The set of candidate options $\mathcal{A}$; Number of bootstraps B .

- 1: Calculate estimated rewards $\hat{\tau}_j$ for all actions $a_j \in \mathcal{A}$ using $\mathcal{D}$
 - 2: Pick the empirical winner $\hat{j}^* \in \arg \max_{j \in \{1, \dots, J\}} \hat{\tau}_j$ that has the largest estimated reward from $\mathcal{D}$
 - 3: **for** $b = 1, \dots, B$ **do**
 - 4: Draw N observations with replacement from $\mathcal{D}$ as a bootstrap sample $\mathcal{D}_b$
 - 5: Calculate bootstrap estimated rewards $\hat{\tau}_{b,j}$ for all actions $j \in \{1, \dots, J\}$ using $\mathcal{D}_b$
 - 6: Pick the bootstrap winner $\hat{j}_b^* \in \arg \max_{j \in \{1, \dots, J\}} \hat{\tau}_{b,j}$ that has the largest estimated reward in the b^{th} bootstrap sample $\mathcal{D}_b$
 - 7: **end for**
 - 8: Calculate the estimated winner's curse $\hat{WC}_{Boot} = B^{-1} \sum_{b=1}^B \hat{\tau}_{b,\hat{j}_b^*} - \hat{\tau}_{\hat{j}^*}$, where $\hat{\tau}_{b,\hat{j}_b^*} - \hat{\tau}_{\hat{j}^*}$ is the difference between the bootstrap estimate of the bootstrap winner and its estimate in the original data
 - 9: **return** $\hat{\tau}_{\hat{j}^*} - \hat{WC}_{Boot}$
-

the index of the action that has the largest estimated reward in the b^{th} bootstrap sample. We call it the bootstrap winner. Following this notation, the largest estimated reward in the b^{th} bootstrap sample is written as $\hat{\tau}_{b,\hat{j}_b^*}$. Then, the bootstrap estimate for the unknown winner's curse WC is:

$$\hat{WC}_{Boot} = \frac{1}{B} \sum_{b=1}^B \hat{\tau}_{b,\hat{j}_b^*} - \hat{\tau}_{\hat{j}^*}.$$

Finally, the correction estimator is computed by plugging the bootstrap-estimated winner's curse back into the bias correction formula in Eq. 4.

Algorithm 1 summarizes the vanilla version of our bootstrap-based approach, which employs a standard nonparametric bootstrap procedure. The standard bootstrap is an easily-implementable method one can apply to a broad range of problems without tuning. As shown later in Section 4 through 6, the standard bootstrap successfully mitigates a substantial portion of the winner's curse across all experiments, though it does not fully eliminate the bias entirely. Two factors limit the finite-sample performance of the standard bootstrap. First, the standard bootstrap tends to underestimate the variance of an estimator (Efron and Gong 1983, Hesterberg 2015), which could lead to under-correction. Second, as we will show below, this bootstrap method converges slowly because the underlying statistic being estimated, the winner's curse, is not Hadamard differentiable. Lemma 4 formalizes this result.

LEMMA 4 (**The Winner’s Curse is Not Fully Hadamard Differentiable**). Define $\mathcal{J}^* = \arg \max_{j \in \{1, \dots, J\}} \tau_j$ as the set of actions with the same highest true expected rewards. Define a realized winner’s curse as $wc(\hat{\tau}; \tau) = \hat{\tau}_{j^*} - \tau_{j^*}$. For any arbitrary direction vector $h \in \mathbb{R}^J$, the directional derivative of the realized winner’s curse $wc(\hat{\tau}; \tau)$ at the point where the estimated reward coincides with the true reward $\hat{\tau} = \tau$ is:

$$wc'_\tau(h) = \sup_{j \in \mathcal{J}^*} h_j.$$

When there is more than one best action, i.e., $|\mathcal{J}^*| \geq 2$, the map $h \mapsto wc'_\tau(h)$ is nonlinear. As a result, the winner’s curse bias $WC = \mathbb{E}_{\mathcal{D}}[wc(\hat{\tau}; \tau)]$ is not fully Hadamard differentiable.

Because the winner’s curse is not fully Hadamard differentiable, Fang and Santos (2019, Theorem 3.1) show that applying the standard bootstrap leads to slow convergence. This slow convergence is particularly problematic in our context, where the winner’s curse is the most pronounced when $\Delta\tau$ is small: even a tiny perturbation in the estimated effects can completely change which action is selected as the winner.

Intuitively, the underperformance of the standard bootstrap stems from the discontinuity of the optimization functional form at points where options are tied. Because each bootstrap sample contains N resampled observations, it closely mimics the original dataset. If one arm happens to win by chance in the original data, it will typically also win in most bootstrap replications. As a result, the bootstrap distribution “inherits” the same sharp kink from the max operator, which leads to slow convergence and an understatement of the uncertainty about which arm is truly best.

Therefore, we apply a subsampling bootstrap method known as the m -out-of- n bootstrap (Bickel et al. 1997), a standard technique for addressing non-Hadamard-differentiable bootstrap problems and mitigating slow convergence. The m -out-of- n bootstrap works by adjusting the resampling scale: rather than drawing bootstrap samples of the same size, N , as the original dataset, it draws smaller bootstrap samples of size $m < N$. This reduced sample size increases the estimation noise in each bootstrap draw, causing the identity of the winner to switch more frequently across bootstrap draws, especially when $\Delta\tau/\sigma$ is small. This additional sampling variability shifts the location of the non-smooth point, and averaging across draws smooths out the non-smooth kink. Figure C.1 in Appendix C.1 provides a visual illustration of how the m -out-of- n bootstrap smooths the kink in a simple two-arm setting.

Therefore, in the m -out-of- n bootstrap, the subsample size m trades off two competing sources of bias. When m is large and close to the original sample size N , the m -out-of- n bootstrap effectively reduces to the standard bootstrap: estimation bias is small, but the discontinuity bias is reintroduced. Conversely, when m is too small, the procedure injects excessive sampling variability that is absent from the original data, inducing substantial downward bias in the estimated winner’s

Algorithm 2 *m*-out-of-*n* Bootstrap Correction

Require: Data $\mathcal{D}$; The set of candidate actions $\mathcal{A}$; Number of bootstraps B .

- 1: Calculate estimated rewards $\hat{\tau}_j$ for all actions $a_j \in \mathcal{A}$ using $\mathcal{D}$
 - 2: Pick the empirical winner $\hat{j}^* \in \arg \max_{j \in \{1, \dots, J\}} \hat{\tau}_j$ that has the largest estimated reward from $\mathcal{D}$
 - 3: **for** $b = 1, \dots, B$ **do**
 - 4: Draw $m = \text{int}(N^{0.95})$ observations with replacement from $\mathcal{D}$ as a bootstrap sample $\mathcal{D}_b$
 - 5: Calculate bootstrap estimated rewards $\hat{\tau}_{b,j}$ for all actions $j \in \{1, \dots, J\}$ using $\mathcal{D}_b$
 - 6: Pick the bootstrap winner $\hat{j}_b^* \in \arg \max_{j \in \{1, \dots, J\}} \hat{\tau}_{b,j}$ that has the largest estimated reward in the b^{th} bootstrap sample $\mathcal{D}_b$
 - 7: **end for**
 - 8: Calculate the estimated winner's curse $\hat{WC}_{Boot} = B^{-1} \sum_{b=1}^B \hat{\tau}_{b,\hat{j}_b^*} - \hat{\tau}_{\hat{j}^*}$, where $\hat{\tau}_{b,\hat{j}_b^*} - \hat{\tau}_{\hat{j}^*}$ is the difference between the bootstrap estimate of the bootstrap winner and its estimate in the original data
 - 9: **return** $\hat{\tau}_{\hat{j}^*} - \hat{WC}_{Boot}$
-

performance and consequently leading to an overcorrection of the winner's curse. This bias trade-off is illustrated in Figure C.2 in Appendix C.1. The right panel shows that the estimation bias decreases monotonically as m increases, whereas the left panel shows that the combined estimation and discontinuity biases first decrease and then increase in m , clearly illustrating the trade-off.

Therefore, m must of small enough to smooth the discontinuity yet large enough to maintain estimation accuracy (Bickel et al. 1997, Bickel and Sakov 2008). In theory, m must grow more slowly than N (formally, $m = o(N)$) for consistency. In practice, a common rule that satisfies this condition is $m = \text{int}(N^{\text{pow}})$ with $\text{pow} \in (0, 1)$. Because the range from $\text{int}(N^{0.9})$ to $\text{int}(N^{0.95})$ provides a reasonable balance between the two biases in a normal A/B testing scenario (see Appendix C.1 for more details), we use the more conservative choice of $m = \text{int}(N^{0.95})$ throughout the paper. We do not change this hyperparameter for different applications to avoid overfitting. The complete procedure of the m -out-of- n bootstrap correction is summarized in Algorithm 2.

3.2. Other Remedies to The Winner's Curse

We identify three main approaches from the current literature to address the winner's curse: Sample splitting, Bayesian methods, and selective inference (also known as post-selection inference). We now discuss each of these in turn.

3.2.1. Sampling Splitting Sample splitting (Cox 1975) addresses the winner's curse by disentangling the decision-making and evaluation process. Since the winner's curse arises from both

picking and evaluating the winner using the same dataset, sample splitting explicitly separates these two stages by dividing the data into two independent subsets: one for decision making, and the other solely for evaluation. Let $\mathcal{D}_{dm}$ and $\mathcal{D}_{eval}$ denote the two subsets used for decision-making and evaluation, respectively. Using only $\mathcal{D}_{dm}$, the firm first estimates the effectiveness of all options and picks a winner, denoted as $\hat{j}_{dm}^*$. Then, using only the evaluation set $\mathcal{D}_{eval}$, the firm estimates the performance of that chosen option, yielding $\hat{\tau}_{\hat{j}_{dm}^*}^{eval}$. Because the evaluation is conducted on a hold-out set that was not involved in the decision selection process, the resulting estimate is free from selection-induced bias, effectively eliminating the winner’s curse. In another word, $\hat{\tau}_{\hat{j}_{dm}^*}^{eval}$ will be the same as $\tau_{j_{dm}^*}$ on average.

However, sample splitting means that both the decision-making and the evaluation are made on smaller samples, which results in two critical issues. First, the data set used to make the decision itself is smaller, meaning that the firms will make incorrect decisions more often. Thus, unlike all other approaches we consider, sample splitting comes at a direct cost to the firm due to the increased probability of making incorrect choices, leading directly to decreased profits. Second, the variance of the estimate of the effectiveness of the chosen option increases drastically with a smaller sample size. This results in wider confidence intervals, which can often be so wide as to cover the full confidence of the other methods that have a small bias, reducing the usefulness of the unbiased estimates. The extent of this issue will be apparent when we report the relative performances of different estimation approaches in Sections 4 through 6.

3.2.2. Bayesian Methods The Bayesian approach involves modeling the data generating process to calculate the posterior mean of the empirically winning action (Efron 2011). Intuitively, the posterior mean reduces the winner’s curse by “shrinking” the Maximum Likelihood Estimation (MLE) estimates towards the prior distribution, given a correctly specified prior.⁴ Because the choice of the prior plays a critical role in the performance of such Bayesian correction (Rasines and Young 2020), researchers often prefer to implement an empirical Bayes approach rather than hand-picking a prior (Efron 2011, Chetty et al. 2018).

Specifically, our paper benchmarks against Efron (2011), a seminal work in applying the empirical Bayes framework to correct for the winner’s curse. This paper shows that when the true performances of all actions are drawn from a common prior with a shared mean, and the observed outcomes under each action are also normally distributed, then the posterior mean of each action’s performance can be computed as the MLE estimate plus a Bayesian correction term:

$$\hat{\tau}_j^{post} = \hat{\tau}_j + \sigma_{MLE} \hat{l}'(\hat{\tau}_j), \forall j \in \{1, \dots, J\}.$$

⁴ We show in Appendix C.3 that a mis-specified prior can instead exacerbate the winner’s curse.

Algorithm 3 Empirical Bayes Correction (Efron 2011)

Require: Dataset $\mathcal{D}$; The set of candidate actions $\mathcal{A}$.

- 1: Compute the MLE estimates $\hat{\tau}_{1:J}$ for all actions $a_j \in \mathcal{A}$ and the sampling standard deviation $\hat{\sigma}_{\text{MLE}}$ using $\mathcal{D}$
 - 2: Pick the empirical winner $\hat{j}^* \in \arg \max_{j \in \{1, \dots, J\}} \hat{\tau}_j$
 - 3: Estimate the log likelihood function $\hat{l}$ of MLE estimates $\hat{\tau}_{1:J}$
 - 4: Compute the posterior mean for the winner $\hat{\tau}_{\hat{j}^*}^{\text{post}} = \hat{\tau}_{\hat{j}^*} + \hat{\sigma}_{\text{MLE}} \hat{l}'(\hat{\tau}_{\hat{j}^*})$
 - 5: **return** $\hat{\tau}_{\hat{j}^*}^{\text{post}}$
-

Here, $\hat{l}'$ is the derivative of the estimated log-likelihood function of the MLEs, $\hat{\tau}$ is the estimated average treatment effect for action j from the MLE estimation, and σ_{MLE} is the standard deviation of the MLE estimate. Efron (2011) assumes a known σ_{MLE} , which is not feasible in reality, so we use the sample standard deviation $\hat{\sigma}_{\text{MLE}}$ as replacement. Algorithm 3 summarizes the empirical Bayes correction procedure.

A limitation for Bayesian approaches that address the winner’s curse is that the models assume that the true effects for each intervention are drawn from a prior distribution with a common mean. This means that these models assume that each treatment has, in expectation, the same effect sizes as the other treatments, as a prior assumption. There are times when such a scenario is feasible. As an example, perhaps one is trying to estimate the influence of background color in affecting the level of sales from an e-commerce website. If you have no theory that any sets of colors may be better than others, then perhaps assuming that the efficacy of each color is drawn from a common normal distribution is reasonable. However, often managers might test treatments they have reasons to believe will outperform the status quo. Examples might include advertising messages that explain benefits better than the status quo advertisements, or new advertisements that are customized based on some economic or psychological theory, or even hunches based on observed patterns. In these cases, one would not expect that the average difference in the effectiveness of such stimuli compared to those not following such optimization to be zero; rather, the marketer would have a theory that one effect is likely better than another. Our experiments reflect this latter philosophy because we assume a fixed difference between the treatments, meaning that the effectiveness of one treatment is distinctly better than the other. In such a case, what the Bayesian correction would then be doing would be to arbitrarily⁵ shrink the assumed differences in the effect sizes of the different treatments. In the following sections we show that using these arbitrary shrinkage factors does indeed reduce the winner’s curse, just like multiplying the fitted differences in performances

⁵ The arbitrariness comes from the fact that the amount of shrinkage is not tied to the correct data generating process.

by a random number less than 1 would, but that these methods are outperformed by the bootstrap methods, where the correction is tied to the distribution of the data.

3.2.3. Selective Inference Selective inference, also known as post-selection inference, addresses the winner’s curse by explicitly deriving the distribution of the winner conditioning on its winning. Such derivation is feasible under certain distributional assumptions and simple optimization problems. In this paper, we focus on one of the most recently developed estimators from Andrews et al. (2024). They found that when selecting a single winner from a discrete group of candidates based on their point estimates, the winner’s original Gaussian estimate becomes truncated normal instead. Then we use the truncated normal distribution to infer the true estimated effect.

Formally, consider J candidate actions $\mathcal{A} = \{a_1, \dots, a_J\}$ with estimated performances $\hat{\tau}_1, \dots, \hat{\tau}_J$. Assume that an estimator $\hat{\tau}_j$ follows a normal distribution $\mathcal{N}(\tau_j, \sigma_j^2)$ for action $a_j \in \mathcal{A}$. Then, conditioning on the event that a_{j^*} has the largest estimate among all candidates, the distribution of $\hat{\tau}_{j^*}$ becomes truncated within an interval $\mathcal{I}$ that depends on the effectiveness of the losing candidates. Let $F_{TN}(\cdot; \mu, \sigma, \mathcal{I})$ denote the cumulative distribution function (CDF) of a truncated normal variable, where the underlying normal distribution has a mean of μ and variance of σ^2 , and the distribution is truncated to lie on the interval $\mathcal{I}$. Note that after truncation, the distribution of the empirical winner’s estimated performance is no longer symmetric. Andrews et al. (2024) proposes that we fix the standard deviation σ_{j^*} from the empirical winner’s original distribution, and move the location μ of the untruncated normal distribution until the observed winner’s value $\hat{\tau}_{j^*}$ sits at the median of the truncated distribution:

$$F_{TN}(\hat{\tau}_{j^*}; \mu, \sigma_{j^*}, \mathcal{I}) = 0.5. \quad (5)$$

In other words, by inverting the definition of the median, the unique solution $\hat{\mu}$ to Eq. (5) is median-unbiased for the winner’s true value τ_{j^*} .

Andrews et al. (2024) provides two types of truncation intervals for $\mathcal{I}$. First, the conditional selective inference estimator uses the second largest estimate $L := \max_{j \in \{1, \dots, J\} \setminus \{j^*\}} \hat{\tau}_j$ as the truncation lower bound, that is, $\mathcal{I} = [L, \infty)$. It follows from the intuition that the winning estimate should be at least as large as the second-largest estimate. However, when $\hat{\tau}_{j^*}$ is only slightly above L (a near-tie), almost all of the truncated mass lies to the right of $\hat{\tau}_{j^*}$. To make $\hat{\tau}_{j^*}$ the truncated median, the estimating equation (Eq. 5) pushes the latent mean μ far left, i.e., a large downward correction.⁶ Andrews et al. (2024) makes a similar observation, where the conditional estimator

⁶ To see this, notice that if the largest two estimates are the same, i.e., $\hat{\tau}_{j^*} = L$, the left hand side of Eq. 5 will be a constant zero and the equation will fail to hold. See Appendix C.4 for a detailed discussion and visual illustration.

yields unreliably wide confidence intervals when the two treatments have very similar levels of effectiveness.

Thus, our paper benchmarks against a second estimator in Andrews et al. (2024), called the hybrid selective inference. In addition to the fact that the winning action should be at least as large as the second best, the hybrid approach constructs a high-probability region, denoted as $[CI_l, CI_u]$, for the winner by combining the confidence intervals from all candidate actions.⁷ Then, the truncation interval used in Eq. 5 is the intersection between this high-probability region and the interval used in the conditional estimator: $\mathcal{I} = [CI_l, CI_u] \cap [L, \infty) = [\max\{CI_l, L\}, CI_u]$. Although this estimator alleviates the wide confidence interval issue from the conditional estimator, we find it does not fix the over-correction issue in near-ties. When the empirical winner $\hat{\tau}_{j^*}$ is close to the truncation lower bound $\max\{CI_l, L\}$, the truncated normal’s feasible mass within the interval $\mathcal{I}$ is still a thin right-tail sliver. Thus, forcing $\hat{\tau}_{j^*}$ to be the truncated median again drives the implied mean far to the left, i.e., overcorrection persists.⁸ Thus, in an actual use case where the signal-to-noise ratio is small, which is precisely when the winner’s curse is severe, the hybrid estimator will reflect a substantial overcorrection.

4. Numerical Experiment: A/B Testing

We now demonstrate how large the winner’s curse can be and evaluate the effectiveness of different fixes in common settings in marketing. We execute this using simulation methods evaluated at empirically realistic parameter values.

We first consider a classic context where a marketer creates a set of different treatments, such as different ad creatives, and decides which one is optimal using using A/B tests. In this section, we consider the case where the consumer response is continuous. In Section 5, we expand the results to allow the consumer responsiveness to vary continuously with customer attributes. Finally, Section 6 examines the winner’s curse in A/B tests with where the consumer response is discrete (e.g., binary purchasing decisions).

4.1. Experimental Setting and Main Results

Consider two candidate treatments, a_1 and a_2 , whose effects on the customer’s response are τ_1 and τ_2 . We assume, without loss of generality, that $\tau_2 \geq \tau_1$. Consistent with the model in Section 2, Customer i ’s response is modeled as a continuous variable following a normal distribution:

$$Y_i = (\tau_1 + \epsilon_{1i})\mathbb{1}\{T_i = a_1\} + (\tau_2 + \epsilon_{2i})\mathbb{1}\{T_i = a_2\}, \quad (6)$$

⁷ This method is called the simultaneous confidence interval approach (Berk et al. 2013) and is referred to as the projection interval in Andrews et al. (2024).

⁸ To see the intuition behind this issue, consider the limit when the winner’s estimate touches the truncation lower bound, i.e., $\hat{\tau}_{j^*} = \max\{CI_l, L\}$. The solution to Eq. 5 would then be $-\infty$.

where $\epsilon_{1i}, \epsilon_{2i} \sim \mathcal{N}(0, \sigma^2)$ are individual-level idiosyncratic shocks. Examples of customer-level continuous responses include spending levels from a catalog or time spent on a platform. For simplicity, we assume that the only consumer heterogeneity comes through the ϵ_{ji} terms.

After drawing N responses for each treatment,⁹ we pick the empirical winner following the inference-then-optimization framework. First, we estimate the effects of each action via simple averaging:

$$\hat{\tau}_j = N^{-1} \sum_{i=1}^{2N} Y_i \mathbb{1}\{T_i = a_j\}, \forall j \in \{1, 2\}.$$

Note that this estimate is both efficient and unbiased. After estimating the treatment effects, we choose the action with the higher estimated effect as the empirical winner:¹⁰

$$\hat{j}^* \in \arg \max_{j \in \{1, 2\}} \hat{\tau}_j.$$

Next, we apply the various estimators we study, including no correction, standard bootstrap, m -out-of- n bootstrap, sample splitting, empirical Bayes, and selective inference, to construct the forecast of the empirical winner’s performance. For a given estimator, the realized winner’s curse in each simulation draw is the difference between the forecasted performance of the empirical winner and its true effect. We repeat this process 1,000 times. We report the average winner’s curse across all draws, denoted as $\bar{WC}$, as a percentage of the actual lift, defined as $\bar{WC}/\Delta\tau$, where $\Delta\tau = \tau_2 - \tau_1$.¹¹ This scaled measure indicates the extent to which the estimated performance improvement exceeds the actual improvement. For example, if the estimated increase in customer spending from sending a promotional email is \$10, but the true increase is \$5, this is reported as a 100% winner’s curse.

Recall from Lemma 2 that two key numbers for designing our simulation experiments are the signal-to-noise ratio, $\Delta\tau/\sigma$, and the number of observations per tested conditions. Because we want to ensure that our simulations provide insight into relevant scenarios, we compile a list of several prominent papers studying personalized targeting (an area where the winner’s curse is likely to be especially high) and A/B testing, and present them in Table 1. The purpose of having this list of papers is not to state that most of the marketing literature has these parameter values, but simply to show that the parameters we examine are relevant to meaningful marketing scenarios.

Given the ranges shown in Table 1, we set the true difference in treatment effects to three different realistic levels, $\Delta\tau \in \{0.005, 0.01, 0.02\}$, where $\sigma = 1$, which translate to signal-to-noise ratios of

⁹ We use 2,500 responses per treatment for most of our analyses.

¹⁰ In many practical contexts, a firm would simply choose the option with the higher point estimate without requiring statistical significance (Gershoff 2018, Feit and Berman 2019, Joo and Chiong 2025).

¹¹ Recall that $\Delta\tau$ is always positive since we assume, without loss of generality, $\tau_2 > \tau_1$. To better understand the shape of the winner’s curse distribution, we also provide the median of the realized winner’s curse across the 1,000 simulation draws for all experiments in this paper in Appendix D.4.

Table 1 A Summary of Signal-to-Noise Ratio and Sample Sizes in Recent Marketing Literature

Paper	Target Variable	Signal-to-Noise Ratio	Dataset Size	Effective Sample Size
Ascarza et al. (2017)	Cell Usage (Minutes)	4.94%	1,041	656 treated and 385 controlled
	Cell Usage (Calls)	5.14%		
	Cell Usage (Texts)	0.76%		
Simester et al. (2020)	Profit	0.02% to 0.20%	3091	$\leq 1,062$
Ellickson et al. (2023)	Purchase Amount	0.28% to 9.19%	1.3M	$\approx 19,118$
Smith et al. (2023)	Profit	0.02% to 3.33%	18,629	≈ 150 (KNN & XGBoost)
Yoganarasimhan et al. (2023)	Subscription Rate	0.28% to 2.79%	337,724	4 to 1651 (Tree-based models)
Daljord et al. (2023)	Length of Stays	3.46%	23,583	16,034 treated and 7,549 controlled
Hitsch et al. (2024)	Spending (2015)	5.88%	292,657	2,475 (Causal KNN)
	Profit (2015)	0.01% to 11.35%		
von Zahn et al. (2025)	Number of Returns	0.2% to 1.96%	78,829	$\leq 39,769$
	Value of Returns	0.18% to 1.21%		
Zheng et al. (2025)	Number of Clicks	1.54%	58,519	29,328 treated and 29,191 controlled
	Order Volume	0.49%		

Effective sample size computes the number of observations for each customer segment-treatment combination. For example, when there are 10,000 observations in total and we equally allocate them to 4 treatments and 25 segments, the effective sample size is 100.

0.5%, 1%, and 2%, respectively. We also select a sample size of $N = 2,500$ per treatment arm, which is on the order of magnitude of the effective sample sizes in several of the reported studies.¹² Note that although datasets in modern marketing applications are becoming larger, and more observations could theoretically alleviate the winner’s curse, Table 1 demonstrates that researchers typically use the larger datasets to analyze a wider range of actions or to capture higher degrees of customer heterogeneity (for example, using machine learning models). As a result, the effective sample size, which describes the number of observations allocated for each customer segment and action, is still small. We demonstrate how the number of observations per treatment changes our results in the next subsection.

Given these realistic parameter levels, we observe a significant winner’s curse, as summarized in Table 2. Looking at the results without correction in the first line, we observe that when the signal-to-noise ratio is 0.5%, the estimated effects overstate the performance improvement by 229%. Even when the signal-to-noise ratio is 2%, the estimated effects remain overestimated by 46%.

¹² When sample sizes differ across treatments, the treatment with the smaller sample size exhibits a larger estimation variance. This inflated variance increases the likelihood that this treatment gets an overly high estimate by chance, thereby amplifying the winner’s curse. We examine this impact of imbalanced designs in detail in Appendix D.2.

Table 2 A/B Testing: Average Winner's Curse $\bar{W}C/\Delta\tau$ Under Different $\Delta\tau/\sigma$

$\Delta\tau/\sigma$	0.5%	1%	2%
No Correction	238.90% (333.83%)	113.85% (171.14%)	47.03% (90.94%)
Standard Bootstrap	82.36% (373.56%)	37.49% (190.41%)	12.37% (100.13%)
m-out-of-n Bootstrap	30.32% (374.77%)	11.65% (191.38%)	-0.32% (101.01%)
Sample Splitting	17.62% (568.95%)	4.87% (283.74%)	2.12% (140.01%)
Empirical Bayes	103.82% (312.38%)	45.25% (161.82%)	10.29% (88.04%)
Hybrid	-75.80%	-39.07%	-14.81%
Selective Inference	(600.11%)	(301.80%)	(144.18%)

Simulation parameters: $\tau_1 = 1$, $\tau_2 = \tau_1 + \Delta\tau$; Response noise level $\sigma = 1$; Sample size $N = 2,500$ per treatment; Results aggregate from 1,000 repetitions; $m = \text{int}(N^{0.95})$ for m -out-of- n bootstrap. Numbers in the parentheses are **standard deviations** of the winner's curse. Sample splitting picks a different winner than the entire sample in 26.1%, 25.5%, and 20.9% of the draws when $\Delta\tau/\sigma$ is 0.5%, 1%, 2%, respectively.

Next, we evaluate the remedies for the winner's curse discussed in Section 3. We begin with our estimator using a standard bootstrap method. As shown in Table 2, our estimator using standard bootstraps significantly mitigates the winner's curse in all cases but does not eliminate the bias. As discussed previously, one reason for the remaining bias is that the underlying process is nonsmooth, causing the standard bootstrap to converge slowly and yield biased estimates of the winner's curse when the sample size is small.¹³ Thus, we consider the estimates from an m -out-of- n bootstrap, which is better suited for the non-smooth estimation. As noted earlier, to avoid overfitting, we always set $m = \text{int}(N^{0.95})$ across our numerical experiments. The m -out-of- n bootstrap performs effectively and further mitigates the winner's curse bias, bringing it down to 21% to -2%.

We compare these bootstrap approaches to the three main approaches that have previously been proposed in the literature. We begin by examining the results from sample splitting. In general, the estimates from sample splitting will have no average bias. While that may seem to solve the winner's curse, there are two important reasons why a firm or researcher may choose not to use sample splitting as their preferred approach. First, the predicted optimal treatments will be wrong more often than they would have been if the optimal treatments had been constructed using

¹³ One can also use a plugin estimator for the winner's curse if there is a closed-form equation for it. In this simple A/B testing setting, we can use the plugin estimator based on the analytical formulation in Lemma 2 to correct the winner's curse. Empirically, this approach achieves performance similar to the standard bootstrap, mitigating the winner's curse but not eliminating it completely (See Appendix D).

the entire dataset. Consequently, when we compare the average performance of the decision from sample splitting to the one constructed from the entire dataset, we find that the decision value of sample splitting loses 2.0%, 3.9%, and 6.9% of the true lift, $\Delta\tau$, when the signal-to-noise ratios are 0.5%, 1%, and 2%, respectively.¹⁴ Thus, unlike the other corrections we consider, the firm must pay for the sample splitting correction through lost profits. Second, since only half of the data is used for evaluation, sample splitting exhibits much larger variance compared to the case without correction. As a result, we observe that while the average estimates from sample splitting are good in this setting, the standard deviations of these estimates are approximately 50% larger than the standard deviation of the bootstrap. Therefore, the precision of the estimate is much smaller as well, so while there is no average bias, the estimated effectiveness that would be obtained by the researcher would likely be much further from the size of the true effect than one would get if one used the m -out-of- n bootstrap instead (over 50% further, on average).¹⁵

Another commonly proposed fix for the winner’s curse is the empirical Bayes estimator by Efron (2011). As discussed in Section 3.2.2, the premise of this estimator, which is common to many empirical Bayes estimators, is that the distributions of effects share a common average level for all the potential treatments. However, this is likely to not fit many real-life interventions, where marketers often want to test the effectiveness of new interventions where the marketer has either a theoretical or an experiential reason for believing that the new intervention is likely to perform better than the status quo. In such a case, the assumption that the two interventions are drawn from a distribution with the same mean would be incorrect. Consistent with this issue, Table 2 reveals that while the empirical Bayes correction reduces the winner’s curse relative to not using a correction, it performs worse than both the standard bootstrap and the m -out-of- n bootstrap.

Finally, we report the winner’s curse using the hybrid selective inference method from Andrews et al. (2024).¹⁶ We observe that this proposed solution over-corrects for the winner’s curse on average, leading to estimates that are further from the true effects than either the standard bootstrap

¹⁴ The decision loss from sample splitting has an inverted U-shape with respect to the signal-to-noise ratios. When the signal-to-noise ratio is zero, doubling the dataset will not improve the decision value. Conversely, when this ratio is very large, a small amount of data is sufficient to pick the right winner, so doubling or halving the dataset does not make a difference. When the ratio is moderate ($\Delta\tau/\sigma = 3.33\%$ when $N = 2,500$ per arm), the decision loss from sample splitting is the largest at 8.3%. See more details in Appendix C.2.

¹⁵ Researchers have also explored alternative splitting strategies, such as allocating the majority of the data (e.g., 90%) for estimation and the remaining 10% of the data for evaluation (Smith et al. 2023). In these cases, the decision correctness increases, but the width of the confidence interval also balloons in size. There are also other ways people can try to modify the sample splitting. For example, one could use a K -fold cross-validation estimator, where the data is partitioned into K subsets, iteratively using $K - 1$ subsets to select the best option (optimization), and evaluating the selected option on the remaining subset. The Jackknife estimator is a special case of this K -fold cross-validation with $K = N$. Despite its conceptual simplicity, the Jackknife estimator does not perform better than sample splitting due to high variance in policy evaluation (See Appendix D for detailed results and discussion).

¹⁶ We also report the results of the conditional inference estimator from Andrews et al. (2024) in Appendix D. The extent of over-correction of the conditional estimator is more severe than the hybrid one.

or especially the m -out-of- n bootstrap. As discussed in Section 3.2.3, such over-correction is the most pronounced when there is a near-tie between the largest two candidates, which is exactly when the winner’s curse is the most severe. Further, the standard deviation of these estimates are much larger than those for the bootstrap approaches, indicating that the “corrected” estimates using this method are much less accurate than what the bootstrap approaches would yield. Finally, note that the selective inference method assumes that the underlying distribution of the estimates are distributed normally, which will happen when the underlying distribution of the randomness is normal. We show in Subsection 4.2 that the selective inference corrections can perform relatively worse than the bootstrap corrections once the underlying variation is no longer normal.

4.2. Impact of the Response Distribution

In this subsection, we consider what happens if the idiosyncratic shocks in the original data-generating process have non-normal shocks. Specifically, we consider that the random shocks could be drawn from the uniform, Laplace, and logistic distributions. We standardize all distributions to have a signal-to-noise ratio of 1%, thereby isolating the impact attributable solely to differences in the shape of the distributions. Table 3 shows that the presence and order of magnitude of the winner’s curse remains largely unaffected by deviations from the normality assumption, ranging from 99.18% to 109.63% when the signal-to-noise ratio is 1%. The correction estimators also have somewhat similar sizes of remaining winner’s curse under the different specifications, although the bootstrap and m -out-of- n bootstrap corrections perform relatively better compared to the other methods when the underlying distributions are not normal. The hybrid selective inference estimator is the most sensitive to the underlying distribution, reflecting the estimator’s reliance on the normality assumption. These selective inference estimates are both very biased, and the standard deviations of the estimates are even larger than they are for sample splitting, meaning that on any particular realization the estimate can be especially bad.

4.3. Impact of the Sample Size

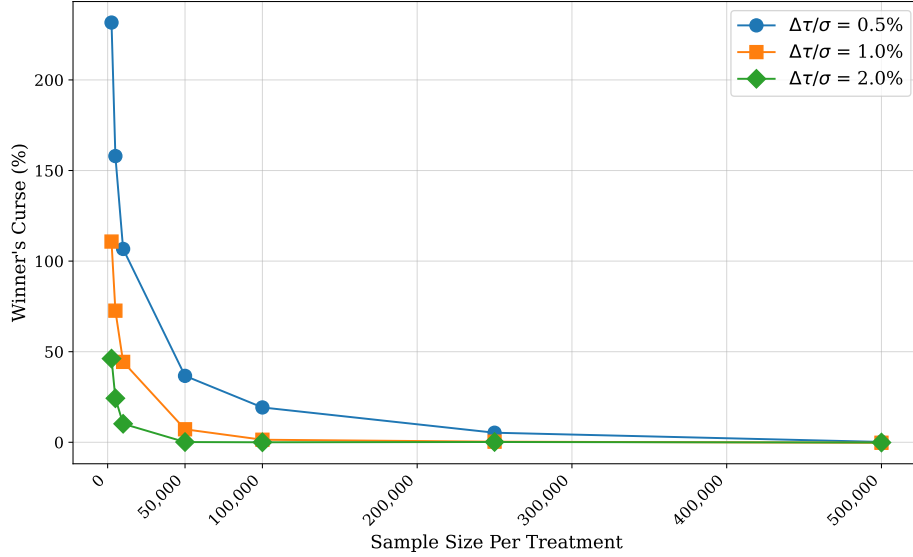
In this subsection, we investigate the impact of the sample size. Although the winner’s curse problem eventually vanishes as we collect a sufficient amount of data, samples used by actual studies are often small enough to have a significant winner’s curse. Further, the number of observations required before the winner’s curse becomes managerially irrelevant depends on the signal-to-noise ratio. Figure 1 shows that when $\Delta\tau/\sigma = 2\%$, we need around $N = 50,000$ observations *per treatment* before the winner’s curse becomes negligible, but the number jumps to around $N = 500,000$ per treatment when the ratio is 0.5%. Given the size of the effects that many academic researchers and companies are trying to tease out (Table 1), we can conclude that the winner’s curse has a meaningful magnitude in many real-life applications.

Table 3 A/B Testing: Average Winner's Curse $\bar{W}C/\Delta\tau$ Under Difference Response Distributions with $\Delta\tau/\sigma = 1\%$

	Normal(0, 1)	Uniform ($-\sqrt{3}, \sqrt{3}$)	Laplace (0, $1/\sqrt{2}$)	Logistic (0, $\sqrt{3}/\pi$)
No Correction	113.85% (171.14%)	108.28% (167.07%)	101.77% (171.10%)	103.99% (170.31%)
Standard Bootstrap	37.49% (190.41%)	33.06% (184.80%)	24.88% (189.80%)	27.58% (188.55%)
m-out-of-n Bootstrap	11.65% (191.38%)	6.86% (185.81%)	-1.07% (190.70%)	1.44% (189.46%)
Sample Splitting	4.87% (283.74%)	-3.65% (275.25%)	-8.58% (275.77%)	-4.91% (276.77%)
Empirical Bayes	45.25% (161.82%)	37.65% (158.12%)	32.72% (162.08%)	34.41% (161.66%)
Hybrid	-39.07% (301.80%)	-32.72% (289.81%)	-47.83% (295.30%)	-43.50% (293.04%)
Selective Inference				

All distributions have variance one. Simulation parameters: $(\tau_1, \tau_2) = (1, 1.01)$; The variance of each distribution is 1; Sample size $N = 2,500$ per treatment; Results aggregate from 1,000 repetitions. $m = \text{int}(N^{0.95})$ for m -out-of- n bootstrap. Numbers in the parentheses are **standard deviations** of the winner's curse.

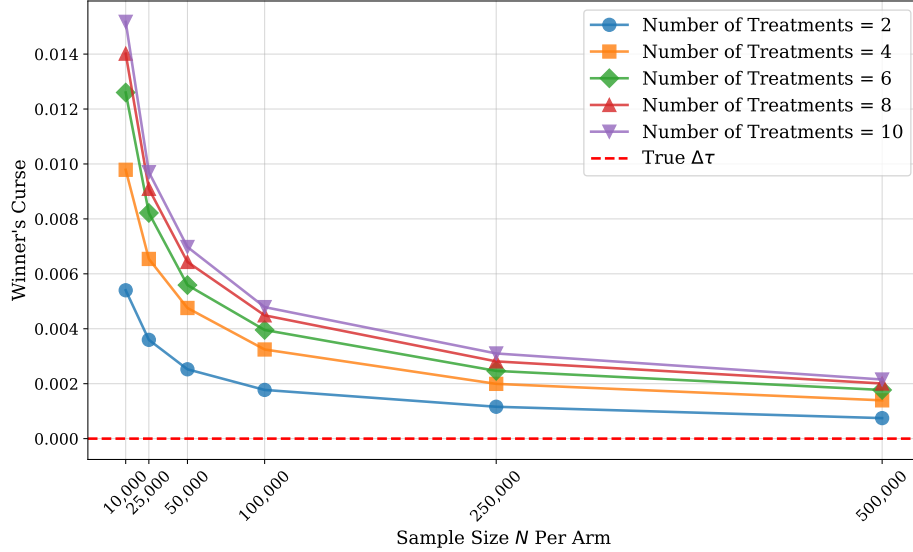
Figure 1 A/B Testing: Winner's Curse $\bar{W}C/\Delta\tau$ Alleviates When Sample Size is Sufficiently Large



Note. Simulation parameters: $\tau_1 = 1$, $\tau_2 = \tau_1 + \Delta\tau$; Response noise level $\sigma = 1$. The different levels of sample size per treatment shown in the figure are: 2.5K, 5K, 10K, 50K, 100K, 250K, and 500K.

Furthermore, when firms have access to large datasets, they typically divide them to study more actions and allow more granular personalization. Thus, as Table 1 shows, a gigantic dataset does not automatically eliminate winner's curse in practice. For example, when firms partition one million customers into 20 segments, there are only 50,000 individuals within each segment, leading

Figure 2 A/B Testing: Sample Size Needed For Insignificant Winner's Curse Increases With the Number of Treatments



Note. Simulation parameters: $\tau_1 = 1$ and the treatment difference $\Delta\tau = 0$; Response volatility $\sigma = 1$; Results aggregate from 1,000 repetitions.

to a significant winner's curse when the signal-to-noise ratio is smaller than 2% (Figure 1). We elaborate more on the implications of increased personalization in Section 5.

4.4. Impact of the Number of Treatments

While most of our analysis focuses on cases where there are two potential actions, we briefly consider how having a greater number of actions available to the marketer affects the magnitude of the winner's curse. In short, the winner's curse becomes more severe and is present for a greater number of observations as the number of choices the marketer tests increases. However, the marginal effect of each additional treatment becomes smaller as the number of treatments increases.

We demonstrate this by assuming that all actions have the same true effects, i.e., $\Delta\tau = 0$, consistent with the theoretical discussion in Lemma 3. This assumption allows us to cleanly isolate the impact of increasing the number of treatments without confounding this effect with different distributions of the differences in the effectiveness of each treatment. We report the winner's curse at its absolute level rather than as percentages of $\Delta\tau$. Figure 2 shows that as the number of candidates increases, the uncorrected estimates of treatment efficacy are increasingly over-optimistic, despite the number of observations per treatment remaining the same. For example, the winner's curse increases about three-fold when the number of candidates increases from two to ten. This increase occurs because when the marketer has more candidates to choose from, the probability of overestimating the effect of at least one option increases.

5. Numerical Experiment: Personalized Targeting

In this section, we consider the winner’s curse in personalized targeting, where a firm customizes treatment assignments for customers with heterogeneous responses to the same intervention. We consider a general setting where the treatment effects vary continuously across customer characteristics. We consider two conditions. In the first condition, the market researcher knows the correct functional form of consumer heterogeneity but not the actual parameters. In the second condition, the true underlying relationship is unknown, so the market researcher relies on machine learning models to estimate these heterogeneous effects and determine the optimal treatment assignment. We show that the winner’s curse not only exists in this setting, but can become more pronounced than in the A/B testing setting, especially when using machine learning models, which introduce additional estimation noise that amplifies the selection bias. Further, we find that our proposed bootstrap methods perform better than the other solutions in these settings.

Consider two candidate treatments, a_1 and a_2 , that the marketer could assign to customers. We assume that the effectiveness of the treatments on a customer’s behavior depends on a one-dimensional standard normal variable, X_i , which summarizes their characteristics. Specifically, we assume that the true effects of each treatment occur through the following process:

$$Y_i = (\tau_1 g(X_i) + \epsilon_{1i}) \mathbb{1}\{T_i = a_1\} + (\tau_2 g(X_i) + \epsilon_{2i}) \mathbb{1}\{T_i = a_2\},$$

where $(\epsilon_{1i}, \epsilon_{2i}) \sim \mathcal{N}(0, \sigma^2)$ are additively-separable idiosyncratic shocks. Note that, in this model, customer heterogeneity is present in both the random shocks, ϵ_{ji} , and the consumer characteristics, through $g(X_i)$.

We parameterize our numerical analysis by setting $g(X_i) = X_i^2 + X_i$, leading to the following data-generating process:

$$Y_i = [\tau_1(X_i^2 + X_i) + \epsilon_{1i}] \mathbb{1}\{T_i = a_1\} + [\tau_2(X_i^2 + X_i) + \epsilon_{2i}] \mathbb{1}\{T_i = a_2\}. \quad (7)$$

We also draw the consumer attributes from a standard normal distribution, which ensures that the average treatment effect across the population is the same as the baseline effects τ_j : $\mathbb{E}_X[\tau_j(X^2 + X)] = \tau_j$.¹⁷ In order to keep our analysis roughly consistent with our analysis of A/B testing, we set $\sigma = 1$, $\tau_1 = 1$, and $\tau_2 = 1.01$, so $\Delta\tau = \tau_2 - \tau_1 = 0.01$, i.e., signal-to-noise ratio is 1%.

We first consider the idealized setting in which the firm knows the true functional form of the consumer heterogeneity but not the parameters. In this case, the firm only needs to estimate the unknown baseline effects, τ_1 and τ_2 . The estimation problem simplifies to the following linear regression:

$$Y_i = (\beta_{11}X_i^2 + \beta_{12}X_i + \epsilon_{1i}) \mathbb{1}\{T_i = a_1\} + (\beta_{21}X_i^2 + \beta_{22}X_i + \epsilon_{2i}) \mathbb{1}\{T_i = a_2\}, \quad (8)$$

¹⁷ We present the results to alternative specifications in Appendix D.3.

where $(\beta_{11}, \beta_{12}, \beta_{21}, \beta_{22})$ are the target parameters to be learned. After learning the parameters, the firm can assign the treatment with the higher predicted effect to each customer. Specifically, for a customer with characteristics X_i , the firm evaluates the predicted effects $\hat{\beta}_{j1}X_i^2 + \hat{\beta}_{j2}X_i$ for each treatment $j \in \{1, 2\}$ and selects the one with the higher prediction.

To evaluate how much the inference-then-optimization procedure overstates the value of this personalized targeting policy, we proceed as follows. For each simulation draw, we first draw $N = 2,500$ training customers per treatment, with $X_i \sim \mathcal{N}(0, 1)$ and outcomes generated from the data generating process in Eq. 7. We then estimate the parameters $(\hat{\beta}_{11}, \hat{\beta}_{12}, \hat{\beta}_{21}, \hat{\beta}_{22})$ in Eq. 8 by ordinary least squares. Next, we draw an independent sample of 10,000 out-of-sample customers, $X_1, \dots, X_{10,000} \sim \mathcal{N}(0, 1)$, which mimics deploying the targeting rule based on our estimates to a new population. For each out-of-sample customer $k \in \{1, \dots, 10,000\}$, we (1) compute the predicted treatment effects

$$\hat{\tau}_j(X_k) = \hat{\beta}_{j1}X_k^2 + \hat{\beta}_{j2}X_k, \forall j \in \{1, 2\},$$

(2) assign the treatment with a higher predicted effect, and (3) compute the corresponding performance forecast for that customer. We then compare the average differences in the estimated vs. actual performance across these 10,000 out-of-sample customers. We repeat this procedure for 1,000 simulation draws and compute the average winner’s curse $\bar{WC}$ across replications. As in Section 4, we report the average winner’s curse $\bar{WC}$ as a percentage of the true lift, $\bar{WC}/\Delta\tau$.

The first column of Table 4, entitled “Correct Functional Form,” shows an average winner’s curse of 88% when we do not utilize any correction. We observe that the bootstrap correction estimator provides the best performance overall, including outperforming the m -out-of- n bootstrap estimator. The degraded performance of the m -out-of- n bootstrap occurs because the treatment effects, $\tau_j g(X_i)$, are now stochastic, varying with different draws of customers, naturally smoothing the non-differentiable point on the value function. Thus, the innate smoothing that the m -out-of- n bootstrap does is of smaller value, while the cost from the increased variation introduced by drawing only a subset m items in each draw leads to degraded performance.

The other proposed corrections also generally perform poorly in this context. Sample splitting has an average winner’s curse that matches the m -out-of- n bootstrap, but again, implementing it would lead to (1) reduced profits by the firm (equal to 4.9% of the lift) and (2) a much larger standard deviation of estimated effects, meaning that while the average size of the effect is reasonable, the estimated effectiveness will often miss with very positive or very negative mistakes. The empirical Bayes correction only corrects for a very small part of the winner’s curse bias, while the selective inference approach over-corrects for the winner’s curse by almost the same magnitude as the original effect itself. Even worse, the selective inference approach has a very large standard deviation across

Table 4 Personalized Targeting: Average Winner's Curse $\bar{WC}/\Delta\tau$ Under Correct Model Specification vs. Causal Forest

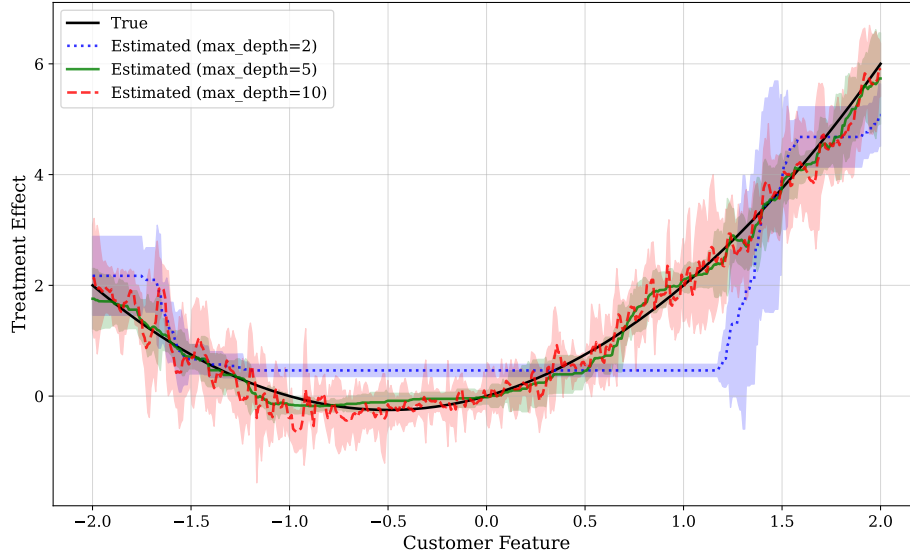
	Correct Functional Form	Causal Forest (Max Depth = 2)	Causal Forest (Max Depth = 5)	Causal Forest (Max Depth = 10)
No Correction	90.00% (106.86%)	230.28% (298.60%)	342.41% (255.87%)	823.10% (253.18%)
Standard Bootstrap	11.40% (122.09%)	89.11% (329.36%)	91.43% (281.33%)	209.41% (275.56%)
m-out-of-n Bootstrap	-19.61% (123.12%)	26.26% (329.37%)	9.70% (282.49%)	108.38% (281.32%)
Sample Splitting	-5.21% (180.93%)	-11.45% (454.06%)	-6.35% (387.22%)	-7.32% (380.41%)
Empirical Bayes	82.11% (106.87%)	-232.54% (318.08%)	120.93% (259.21%)	564.45% (257.05%)
Hybrid	-63.95%	-1906.50%	-1425.85%	-1554.43%
Selective Inference	(192.21%)	(749.95%)	(500.32%)	(355.24%)

Simulation parameters: $(\tau_1, \tau_2) = (1, 1.01)$; Response noise level $\sigma = 1$ so the signal-to-noise ratio $\Delta/\sigma = 1\%$; $g = x^2 + x$; Sample size is $N = 2,500$ per treatment; Results aggregate from 1,000 repetitions; $m = \text{int}(N^{0.95})$ for m -out-of- n bootstrap. Numbers in the parentheses are **standard deviations** of the winner's curse. Sample splitting picks a different winner than the entire sample in 27.93%, 27.89%, and 29.50% of the draws when using a causal forest with maximum depth of 2, 5, and 10, respectively. For the correct functional form, sample splitting's winner differs in 24.30% of the draws.

the 1000 draws, reflecting that on the one draw that a researcher would observe in reality is likely to be very far from the true accuracy.

We next consider a more realistic setting where the firms do not know the true functional form of the data generation process. Many firms and researchers instead turn towards using machine learning methods to estimate the underlying heterogeneity (Rafieian and Yoganarasimhan 2023, Yoganarasimhan et al. 2023, Hitsch et al. 2024). One of the most commonly used models by both researchers and practitioners is the causal forest model (Wager and Athey 2018), partially because these models enjoy nice theoretical guarantees such as consistency.

A key tuning parameter in causal forests is the maximum tree depth. The maximum depth controls model flexibility by changing how finely the model partitions the covariate space. For example, a maximum depth of five means that each tree within a causal forest model independently approximates customer heterogeneity by grouping the customers into at most $2^5 = 32$ segments and then estimating the treatment effects segment by segment. Increasing the depth can reduce the approximation bias by fitting a richer pattern of customer heterogeneity, but more flexibility also brings a larger estimation variance. Figure 3 shows that when the maximum depth increases from two to five, the causal forest can approximate the true heterogeneous effects better. But further increasing the model flexibility yields only marginal improvement on approximation, but significantly larger variation. In practice, as computing power has improved and larger datasets

Figure 3 Personalized Targeting: Causal Forest with Different Depths

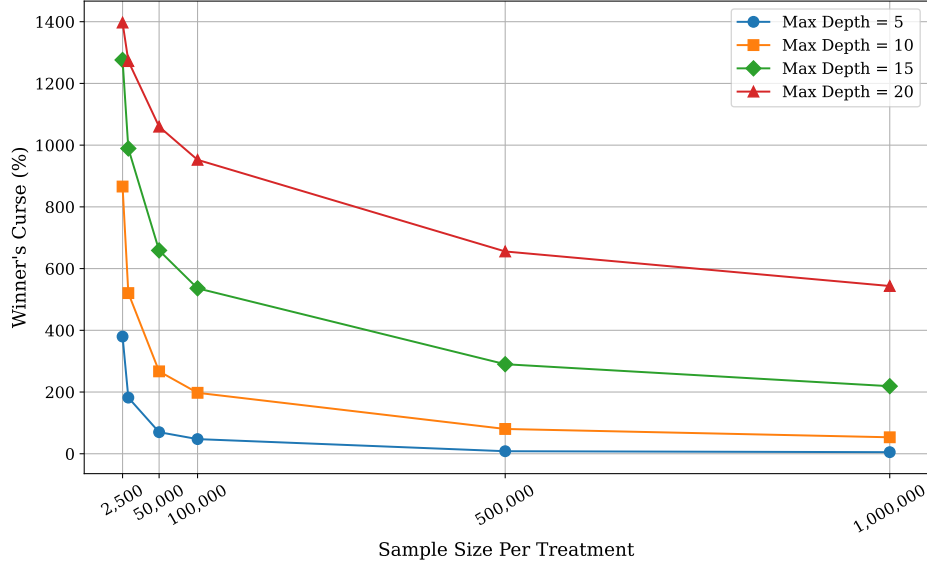
Note. Simulation parameters: Sample size is $N = 2,500$; $\tau = 1$, $g(x) = x^2 + x$; The heterogeneous effect is then $\tau(X_i) = X_i^2 + X_i$. The black curve represents this true heterogeneous effect. We fit three causal forests using one draw from the data generation process. Each causal forest consists of 100 causal trees. The blue dotted line, green solid line, and the red dashed line represents causal forest models with maximum depth 2, 5, and 10, respectively.

have become available, firms and researchers typically leverage these resources to deploy more flexible models to approximate the underlying data generation process better. However, the growing variance accompanied with this growing flexibility inevitably means a more severe overestimation in the performance forecast. To quantify this impact, we examine the impact of the different maximum tree depths in the personalized targeting application.

The causal forest simulations are essentially the same as described above. After drawing $N = 2,500$ observations for each treatment, we train a causal forest model on this training data. Then, for each of the 10,000 out-of-sample customers, we use the trained forest model to predict the effects of each treatment and assign the one with the higher predicted effect. Finally, we compare the estimated effectiveness of the assigned treatments to their corresponding true values to compute the winner’s curse. Repeating this procedure for 1,000 draws yields the average winner’s curse relative to the true lift reported in the “Causal Forest” columns in Table 4.

The “No Correction” results in these columns show that the estimated lifts from the personal treatment assignments based on the causal forest estimates are severely inflated. This overestimation occurs because the flexible functional form of the causal forest means that the effective number of observations per leaf will be smaller. The impact from such a reduction in the effective sample size is more pronounced for customers whose attributes, X_i , fall in ranges where there are fewer observations, since overfitting can be especially pronounced in this region. That said, if X_i is

Figure 4 Personalized Targeting: Winner’s Curse Does Not Improve Significantly with Sample Size If We Use Highly Capable Causal Forests



Note. Simulation parameters: $(\tau_1, \tau_2) = (1, 1.01)$, $g = x^2 + x$; Response volatility $\sigma = 1$; 10,000 out-of-sample customers; Results aggregate from 1,000 repetitions. The different levels of sample size per treatment shown in the figure are: 2.5K, 10K, 50K, 100K, 500K, 1M.

very positive or very negative then the winning treatment will be enough ahead that luck cannot generally change which treatment ends up as the winner.

The winner’s curse worsens when the firm further increases the model flexibility for better personalization. Table 4 shows that when we increase the maximum depth from two to ten, the winner’s curse without any correction increases from 231.37% to 831.19%. Even worse, increasing the sample size is not an easy fix for this increasing overestimation. Figure 4 shows that if the firm uses a causal forest with a maximum depth of 15, the winner’s curse remains above 200% even when the researcher has one million observations per treatment. The reason is that a 15-layer forest can, at most, partition customers into almost 32 thousand possible segments, leaving only a handful of observations in each.

Conversely, the bootstrap-based methods, especially the m -out-of- n bootstrap, provides a relatively good fix for the winner’s curse.¹⁸ As before, sample splitting performs reasonably on average, but (1) a firm that uses sample splitting will earn lower profits (equal to 2.1% of the lift when using a maximum depth of five for the causal forest),¹⁹ and (2) it has a much wider standard deviation, reflecting that sample splitting provides especially unreliable estimates. The empirical Bayes

¹⁸ Because tree models are inherently non-smooth due to branching, nonsmoothness now comes from both selecting the winner and the causal forest itself. Hence, the same level of smoothing ($N = \text{int}(N^{0.95})$) works better for the m -out-of- n bootstrap compared to the case with a correct functional form.

¹⁹ Similar to the discussion in Section 4.1, the performance gap between sample splitting and using the full dataset exhibits an inverted-U shape on the difficulty of the learning problem. For example, when we increase the sample size

and hybrid selective inference estimators perform much worse than the m -out-of- n bootstrap. The selective inference estimator performs especially poorly. This poor performance arises because with personalized targeting there will always be some set of customers where the true impact of both actions are very close. We already noted that when the true effects are very close, the selective inference estimators have a huge negative bias, and this negative bias is not counterbalanced by a positive balance for customers where the impact of the two effects is far apart. We also note that despite the median-unbiased design of the selective inference approach, the median winner’s curse across the 1,000 repeated experiments would still be over-corrected at -1374.75% for a causal forest with maximum depth of five.

6. Numerical Experiment: Binary Outcomes

Thus far, our analysis has focused on marketing actions that have continuous outcomes, such as consumer spending or consumer app usage. However, in many cases there might be binary outcomes. For example, many advertisements promote a specific product, and the customer either buys or does not buy that product. In other scenarios, the goal of an advertisement might simply be to induce a click on an ad, and the rest of the customer journey might not even be observable to the marketing researcher. Moreover, some companies only have one product, or are trying to get customers to take a specific action.

We first revisit the A/B testing setting with a single customer segment, as in Section 4. To model binary outcomes, we assume a Bernoulli data generation process:

$$\Pr(Y_i = 1) = \tau_1 \mathbb{1}\{T_i = a_1\} + \tau_2 \mathbb{1}\{T_i = a_2\}.$$

In this specification, each treatment a_j produces a positive outcome, such as purchase or a click, with probability τ_j . We set the baseline success probability $\tau_1 \in \{0.1, 0.2, 0.3\}$, and for each baseline τ_1 , we consider two treatment lifts $\Delta\tau = \tau_2 - \tau_1$ of 0.001 and 0.005. This design allows us to examine how both the size of the extra lift and the baseline success rates affect the winner’s curse.

Table 5 shows that the core pattern from the continuous-outcome setting carries over: without correction, the winner’s curse is substantial and becomes more severe as the signal $\Delta\tau$ shrinks. However, the magnitude of the winner’s curse for binary outcomes also depends on the absolute level of the success probability τ_j , and not just $\Delta\tau$. For a Bernoulli outcome, the variance $\text{Var}(Y_i|T_i = a_j) = \tau_j(1 - \tau_j)$ is the largest when τ_j is 0.5, and is smaller when the outcome is very rare or very common. Holding $\Delta\tau$ fixed, moving the baseline rate from 0.1 to 0.3 raises the outcome variance, and therefore makes it harder to distinguish between the two treatments. As established throughout the paper, higher variances lead to a larger winner’s curse. Consequently, for the same $\Delta\tau$, the upward bias is greater when the baseline success probability moves from 0.1 toward 0.3.

from $N = 2,500$ to $N = 10,000$, which in turn reduces the uncertainty, the sample splitting’s loss of profit increases from 2.1% to 4.6% of the true lift for a causal forest with maximum depth five.

Table 5 Binary Outcomes in A/B Testing: Average Winner's Curse $\bar{W}C/\Delta\tau$ Under Different Effect Sizes

(τ_1, τ_2)	$\Delta\tau = 0.001$			$\Delta\tau = 0.005$		
	(0.1, 0.101)	(0.2, 0.201)	(0.3, 0.301)	(0.1, 0.105)	(0.2, 0.205)	(0.3, 0.305)
No Correction	341.86% (505.39%)	433.96% (697.34%)	530.70% (788.21%)	59.75% (108.48%)	79.60% (145.30%)	98.61% (163.20%)
Standard Bootstrap	103.14% (558.96%)	118.25% (770.07%)	175.50% (871.87%)	15.24% (119.11%)	18.32% (159.48%)	28.87% (179.69%)
m-out-of-n Bootstrap	24.64% (558.78%)	11.62% (771.70%)	54.09% (874.18%)	-0.51% (119.53%)	-2.80% (160.30%)	4.77% (180.45%)
Sample Splitting	-17.10% (836.94%)	-42.92% (1147.04%)	-14.98% (1354.17%)	-0.78% (171.19%)	-8.65% (231.85%)	-6.91% (276.63%)
Empirical Bayes	134.60% (465.73%)	158.84% (646.76%)	208.99% (741.39%)	16.07% (102.88%)	22.84% (137.58%)	35.05% (154.49%)
Hybrid	-95.75% (868.00%)	-155.54% (1192.75%)	-115.85% (1322.81%)	-19.29% (174.38%)	-31.74% (236.74%)	-35.71% (277.33%)
Selective Inference						

Simulation parameters: $\tau_2 = \tau_1 + \Delta\tau$; Sample size $N = 2,500$ per treatment; Results aggregate from 1,000 repetitions; $m = \text{int}(N^{0.95})$ for m -out-of- n bootstrap. Numbers in the parentheses are **standard deviations** of the winner's curse.

The performance of the correction estimators in Table 5 follows the same pattern as in the continuous-outcome experiments. The standard bootstrap reduces the relative winner's curse significantly, but the m -out-of- n bootstrap still performs the best. As before, sample splitting will not have a theoretical bias, but it suffers from (1) using sample splitting would lead the firm to make poorer choices, which would lead to lost profits²⁰, and (2) because sample splitting uses only half the sample, it is much more random, meaning that while it is better on average, it may not be better for any particular realization. In fact, in our case with 1000 random simulations, the randomness was so high that by happenstance the average in Table 5 for sample splitting is occasionally higher than the average for the m -out-of- n bootstrap. The empirical Bayes mitigates some of the winner's curse, but still leaves behind a substantial portion of the winner's curse bias. The hybrid selective inference approach severely over-corrects the bias and has a large variance across draws, meaning that this approach is especially problematic and especially unlikely to deliver accurate estimates.

We next turn to personalized targeting with binary outcomes. We retain the Bernoulli structure but now allow treatment effects to vary with customer characteristics X_i , analogously to the heterogeneous-effects setting in Section 5:

$$\Pr(Y_i = 1) = \tau_1 g(X_i) \mathbb{1}\{T_i = a_1\} + \tau_2 g(X_i) \mathbb{1}\{T_i = a_2\}.$$

To ensure that the purchase probability falls within zero and one, we set $X_i \sim \text{Uniform}(0, 1)$. Recall that $g(X_i)$ captures how observable characteristics shift purchase probabilities under each

²⁰ For example, when customers' purchase probabilities under the two treatments are $(\tau_1, \tau_2) = (0.1, 0.105)$, sample splitting leads to losses equal to 4.72% of the actual lift compared to using the entire dataset.

Table 6 Binary Outcomes in Personalized Targeting: Average Winner's Curse $\bar{W}C/\Delta\tau$ Under Different Effect Sizes

(τ_1, τ_2)	$\Delta\tau = 0.001$			$\Delta\tau = 0.005$		
	(0.1, 0.101)	(0.2, 0.201)	(0.3, 0.301)	(0.1, 0.105)	(0.2, 0.205)	(0.3, 0.305)
No Correction	909.72% (463.15%)	1210.57% (575.19%)	1392.26% (640.57%)	178.94% (92.89%)	239.91% (117.12%)	276.99% (128.69%)
Standard Bootstrap	179.14% (490.30%)	184.06% (625.01%)	229.04% (688.18%)	32.43% (98.21%)	36.69% (127.78%)	41.68% (136.84%)
m-out-of-n Bootstrap	-35.36% (501.27%)	-80.86% (622.70%)	-87.04% (694.60%)	-8.85% (100.20%)	-19.20% (124.52%)	-18.30% (137.96%)
Sample Splitting	-1.25% (655.49%)	-41.08% (866.77%)	-22.87% (999.84%)	-1.73% (131.89%)	-5.14% (173.15%)	-0.90% (200.87%)
Empirical Bayes	283.25% (461.60%)	643.89% (584.01%)	978.16% (652.93%)	53.68% (92.78%)	127.16% (119.29%)	194.44% (131.05%)
Hybrid	-845.76%	-1219.39%	-1446.99%	-174.59%	-243.51%	-286.97%
Selective Inference	(636.80%)	(826.61%)	(962.41%)	(128.91%)	(170.84%)	(192.32%)

Simulation parameters: $\tau_2 = \tau_1 + \Delta\tau$; Sample size $N = 2,500$ per treatment; Results aggregate from 1,000 repetitions; $m = \text{int}(N^{0.95})$ for m -out-of- n bootstrap. Numbers in the parentheses are **standard deviations** of the winner's curse.

treatment and we set $g(X_i) = X_i^2 + X_i$ as before. We estimate the heterogeneous effects via a causal forest with maximum depth of five. We follow the same simulation procedure as in Simulation 2, now with binary outcomes.

Table 6 shows that personalization magnifies the winner's curse in the binary setting as well. Among the correction methods, the m -out-of- n bootstrap delivers the best correction performance on average, with the exception of sample splitting, which as we observed before, has no average bias but (1) comes at a cost of profit losses for the firm, and (2) has a larger standard deviation of its estimates across the draws, so it is as likely to deliver an estimate on any particular draw that is further away from the truth than the m -out-of- n bootstrap would deliver.

Taken together, these experiments demonstrate that our main conclusions are not specific to continuous outcomes. In realistic binary-outcome environments, where firms care about conversion rates rather than spending levels, the winner's curse can be large, particularly when success probabilities are moderate and treatment effects are small. We demonstrate that the m -out-of- n bootstrap performs well across both the simple A/B test and the personalized targeting policy contexts, demonstrating the method's robustness.

7. Conclusion

In this paper, we show that the winner's curse, a systematic over-optimism about the value of the chosen action, arises naturally in a wide range of data-driven marketing decisions that follow an inference-then-optimization framework. Because firms select actions based on noisy estimates, they tend to implement actions whose effectiveness is over-estimated. We document that such inflated

evaluation can be large in realistic A/B testing and personalized targeting settings, especially when the signal-to-noise ratio is low, when many actions are tested, or when firms use flexible machine learning models for personalization. As firms increasingly rely on flexible learning models and evaluate massive pools of candidate options, they should expect systematic over-optimism in estimated performance unless they explicitly correct for the winner’s curse. We propose an easy-to-implement bootstrap-based correction that generally performs well at mitigating a large fraction of the winner’s curse. We compare the performance of our bootstrap approach against three of the most common existing remedies: sample splitting, empirical Bayes, and selective inference. We show that each of the existing solutions has important limitations, and they tend to perform poorly in realistic marketing contexts in comparison to the bootstrap estimator.

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ONLINE APPENDIX

A. Literature Parameters

We demonstrate how we calculate the signal-to-noise ratios in Table 1. We will articulate the source of data and parameters that we use to compute the signal-to-noise ratio $\Delta\tau/\sigma$ for each paper. In general, $\Delta\tau$, the difference between options, is more clearly documented in papers. Standard deviation σ , on the other hand, is sometimes not reported. In those cases, we infer the standard deviation using the sample size N and standard error SE via $\sigma = SE * \sqrt{N}$.

Ascarza et al. (2017) investigates the spillover effect from targeted customers via social networks in the telecommunication industry. Through an A/B test where the two arms are target or not target, the authors found that the cellphone usage of the targeted customers decreases more slowly compared to the control group. Specifically, the improvement measured in minutes is $\Delta\tau = 6.54$ (Table 4), the standard deviation is 132.4 (Table 2), giving a signal-to-noise ratio of 4.94%. Similarly, the signal-to-noise ratio for improvement measured in the number of calls is $3.40/66.2 = 5.14\%$, and for improvement measured in the number of texts is $1.68/222 = 0.76\%$. 1,041 customers participated in the experiment, with 63% of them (656) in the treated group and 37% (385) in the control group.

Simester et al. (2020) proposes a more efficient method for evaluating and comparing marketing targeting policies. Their dataset contains 3,091 carrier routes, and the dataset is divided into two different treatment conditions and one control group, with the control group having the most observations (1,062 from Table 2). There are three new policies comparing with the standard lasso policy, as documented in Table 5. The maximum difference between two arms is 0.028, while the minimum difference is 0.003. The standard deviation, on the other hand, is less clear. Because the paper only provides the standard error for the difference between two policies rather than the standard error for just one policy, we can only infer the range of the standard deviation. Based on the results in Table 4 and the sample sizes in Table 3, the standard deviation ranges from \$9 to \$19, with the RBP design having the largest volatility. We use the average, which is \$14 as the standard deviation. The resulting signal-to-noise ratio is then between 0.02% to 0.20%.

Ellickson et al. (2023) studies the causal effects of different targeted email promotions on the decisions of consumers to open an email and make a purchase. Based on Table 1 in the appendix, there are 34 candidate emails including the baseline. We look at the purchase amount, and we see that the maximum signal-to-noise ratio is between Email 1 and Email 5 at $(0.51 - 0.01)/\sqrt{(1.26^2 + 7.59^2)/2} = 9.19\%$. As for the minimum ratio, because the numbers reported are rounded to two decimal places, there is more than one emails yields the same purchase amount at \$0.11. Thus, we know that the minimum signal is less than \$0.01 and we can compute the upper

bound of the minimum signal-to-noise ratio is around 0.28%. This paper has a dataset with 1.3 million customers, which are divided into 34 treatment groups, each with a different email promotion. Additionally, they identify two customer segments using the K-Means algorithm. Thus, the effective sample size per treatment per segment is $1.3M/34/2 \approx 19,118$.

Smith et al. (2023) studies the profitability of personalized pricing by comparing different machine learning and econometric models. After data cleaning, their targeting policies are constructed on a dataset with 18,629 purchase occasions. Their Table 4 reports the profitability of different machine learning models. Each model-features combination is a separate arm in the experiment. For example, the $\Delta\tau$ of the Bayesian hierarchical logit model with normal heterogeneity when using Base Demos only compared with blanket targeting is $7.53 - 6.45 = 1.08$. The approximate standard deviation is $\sqrt{18629} \cdot \sqrt{(0.47^2 + 0.32^2)/2} = 54.88$. Together, the approximate signal-to-noise ratio is $\Delta\tau/\sigma = 1.97\%$. Similarly, we can calculate the ratios for every model-features combination. The maximum signal-to-noise ratio can be computed at 3.33%. Note that the numbers reported are rounded to two decimal places, there is more than one emails yields the same profit at \$5.50. Thus, we know that the minimum signal is less than \$0.01 and we can compute the upper bound of the minimum signal-to-noise ratio is around 0.02%. Lastly, we can infer the effective sample size per segment and treatment when a segmentation is clear. For example, when using an XGBoost model, each purchase occasion is put into one end node (leaf) of a tree, effectively providing a segmentation for the observations, similar to the causal forest model. Because the trees in the XGBoost model have at least a maximum depth of six according to Table C.1, the effective sample size per leaf is about $18629/2^6/2 \approx 146$ (two price levels). The K Nearest Neighbor (KNN) method also enjoys interpretable customer segmentation, based on which we can infer that the effective sample size is at most $301/2 \approx 151$.

Yoganarasimhan et al. (2023) uses data from a large-scale field experiment to study the effect of free trial length on customer acquisition, retention, and profitability, and then develops and evaluates a personalized policy for assigning optimal trial lengths to individual users. One of the important metrics for evaluation is the lift in subscription rate. The standard deviation of it is 35.5% (Table 3). The candidate options under selection are personalized policies with different machine learning models, and the uniform targeting policies with different free trial-lengths. The maximum signal-to-noise ratio attains between the Lasso-based personalized policy and the uniform policy with 30-day free trial at $\Delta\tau/\sigma = (15.62\% - 14.63\%)/35.5\% = 2.79\%$. On the other hand, the minimum signal is less than 0.01% between the CART-based and the causal tree-based policy. This gives an upper bound on the minimum ratio at $0.01\%/35.5\% = 0.28\%$. As for the effective sample size per treatment-segment combination, because the paper uses 70% of the 337,724 observations for training, the training set size is 236,407. Then, we can infer the effective sample size per

end node (leaf) from the hyperparameters provided in the appendix. For example, the minimum number of treatment and control observations in the causal tree is from 100 to 900. The number for causal forest is from 4 to 1651. Their XGBoost model did not specify the minimum number of observations in a leaf, but rather specified the maximum depth, like the setting in our paper. The maximum depth used in Yoganarasimhan et al. (2023) is 12, so the effective sample size in each leaf is $236407/2^{12} \approx 58$.

Daljord et al. (2023) outlines an experiment-based approach using Marginal Treatment Effects (MTEs) to optimally design and target compliance promotions. They documented in Section 2.2.2 that when across different experiment blocks, the average stays are 0.09 days higher than the baseline tier. The standard deviation of stays across all samples is 2.60. Together, we can compute the signal-to-noise ratio as $0.09/2.60 = 3.46\%$.

Hitsch et al. (2024) presents and evaluates a framework for creating profitable customer targeting policies. On the ATE level, the lift in spending in 2015 is 2.561 (Table 2), while the standard deviation is 43.549 (Table 1), resulting in a signal-to-noise ratio of 5.88%. For the targeting policies, Table 5 reports the profits from different targeting models as well as the bootstrap standard error. We treat different targeting models and blanket mailing as the set of candidate options. The maximum signal-to-noise ratio attains between Causal Forest model and blanket mailing at $\Delta\tau/\sigma = 265/\sqrt{1000 \cdot (63.8^2 + 82.7^2)/2} \approx 11.35\%$. The standard deviation is inferred by multiplying the bootstrap standard error by the square root of the number of bootstraps. The minimum ratio attains between Random Forest and TOR (Lasso) models at $\Delta\tau/\sigma = (2356.2 - 2355.9)/\sqrt{1000 \cdot (91.6^2 + 98.4^2)/2} \approx 0.01\%$

von Zahn et al. (2025) uses an A/B test to examine the impact of green nudging on product returns. The results of this experiment are in Table 2. There are three arms in the experiments: control, single nudge, and dual nudge. For the number of return shipments, the maximum signal-to-noise ratio attains between single nudge and dual nudge at $\Delta\tau/\sigma = (0.2608 - 0.253)/\sqrt{(39060 \cdot 0.002^2 + 39769 \cdot 0.002^2)/2} \approx 1.96\%$. The minimum ratio attains between single nudge and control group at 0.2%. Similarly, for the value of return shipments, the maximum signal-to-noise ratio attains between dual nudge and control arm at $\Delta\tau/\sigma = (18.49 - 17.88)/\sqrt{(0.26^2 \cdot 38475 + 0.25^2 \cdot 39769)/2} \approx 1.21\%$. The minimum ratio attains between single nudge and control arm at 0.18%.

Zheng et al. (2025) uses an A/B test to examine how query recommendation improves clicks, order volumes, and other metrics. It increases the number of clicks (Tap Volume in Table 2) by $\Delta\tau = 0.388$. The average standard deviation across the treated and control groups, based on the standard error provided in Table 1 and the sample sizes, is $\sigma = \sqrt{(0.133^2 \cdot 29328 + 0.160^2 \cdot 29191)/2} \approx 25.16$. The signal-to-noise ratio is then $\Delta\tau/\sigma = 1.54\%$. Another important metric reported in the paper

is the order volume, which is increased by $\Delta\tau = 0.036$ with the query recommender system. The average standard deviation is $\sigma = \sqrt{(0.043^2 * 29328 + 0.043^2 * 29191)/2} \approx 7.36$, which gives a signal-to-noise ratio of $\Delta\tau/\sigma = 0.49\%$.

B. Proofs

LEMMA B.1 (Convexity of a max function). *Let $J \in \mathbb{N}$ be the fixed number of candidate options, and each option is denoted as $x_j \in \mathbb{R}$ for $j \in \{1, \dots, J\}$. The max mapping $f: \mathbb{R}^J \mapsto \mathbb{R}$ defined as*

$$f(x_{1:J}) = \max_{1 \leq j \leq J} x_j,$$

is a convex function.

Proof of Lemma B.1 Consider arbitrary $x, y \in \mathbb{R}^J$ and $\lambda \in [0, 1]$. For every element $j \in \{1, \dots, J\}$, we have:

$$\lambda x_j + (1 - \lambda)y_j \leq \lambda \max_{1 \leq k \leq J} x_k + (1 - \lambda) \max_{1 \leq k \leq J} y_k = \lambda f(x) + (1 - \lambda)f(y).$$

Because this inequality holds for every j , we have:

$$f(\lambda x + (1 - \lambda)y) = \max_{1 \leq j \leq J} \{\lambda x_j + (1 - \lambda)y_j\} \leq \lambda f(x) + (1 - \lambda)f(y).$$

This is precisely the definition of a convex function. Thus, f is convex. Q.E.D.

Proof of Lemma 1 Consider the expected value of the empirical winner's value:

$$\mathbb{E}[\hat{\tau}_{j^*}] = \mathbb{E}[\max_{1 \leq j \leq J} \hat{\tau}_j] \geq \max_{1 \leq j \leq J} \mathbb{E}[\hat{\tau}_j].$$

The inequality holds because of Jensen's inequality and the convexity of the maximum function (Lemma B.1). Because $\hat{\tau}_i$ are unbiased estimators for the true parameters τ_i , we have $\max_{1 \leq j \leq J} \mathbb{E}[\hat{\tau}_j] = \max_{1 \leq j \leq J} \tau_j$. Finally, for any realization of the empirical winner $\hat{j}^*$, we have $\max_{1 \leq j \leq J} \tau_j \geq \tau_{\hat{j}^*}$. Thus, we get the winner's curse conclusion in Eq 2:

$$\mathbb{E}[\hat{\tau}_{j^*}] \geq \mathbb{E}[\tau_{j^*}]$$

Q.E.D.

Proof of Lemma 2 Decompose the estimators for each option as $\hat{\tau}_j = \tau_j + \xi_j$ with $\xi_j \sim \mathcal{N}(0, \sigma^2/N)$ for $j \in \{1, 2\}$. Write the winner's curse as follows:

$$WC = \mathbb{E}[\hat{\tau}_{j^*} - \tau_{j^*}] = \mathbb{E}[\xi_1 \mathbb{1}\{\xi_1 - \xi_2 \geq \tau_2 - \tau_1\}] + \mathbb{E}[\xi_2 \mathbb{1}\{\xi_2 - \xi_1 \geq \tau_1 - \tau_2\}]. \quad (\text{B.1})$$

Consider the first term $\mathbb{E}[\xi_1 \mathbb{1}\{\xi_1 - \xi_2 \geq \tau_2 - \tau_1\}]$. Orthogonalizing ξ_1, ξ_2 by defining:

$$U = \frac{\xi_1 - \xi_2}{\sqrt{2}\sigma/\sqrt{N}}, \quad V = \frac{\xi_1 + \xi_2}{\sqrt{2}\sigma/\sqrt{N}}.$$

U and V are independent standard normal variables. Then,

$$\mathbb{E}[\xi_1 \mathbb{1}\{\xi_1 - \xi_2 \geq \tau_2 - \tau_1\}] = \frac{\sigma}{2} \sqrt{\frac{2}{N}} \varphi\left(\frac{\Delta\tau}{\sqrt{2}\sigma/\sqrt{N}}\right).$$

Because the two terms in Eq. B.1 are symmetric, we have:

$$WC = \sigma \sqrt{\frac{2}{N}} \varphi\left(\frac{\Delta\tau}{\sigma} \sqrt{\frac{N}{2}}\right).$$

Q.E.D.

Proof of Lemma 3 The proof follows the logic of Leadbetter et al. (1983).

Decompose the estimators for each option as $\hat{\tau}_j = \tau_j + \xi_j$, where $\xi_j \sim \mathcal{N}(0, \sigma^2/N)$. The winner's curse can be written as:

$$WC = \mathbb{E}[\hat{\tau}_{j^*} - \tau_{j^*}] = \mathbb{E}[\xi_{j^*}] = \frac{\sigma^2}{N} \mathbb{E}[Z_{(J)}].$$

$Z_{(J)}$ represents the maximum of J i.i.d. standard normal variables $\{Z_1, \dots, Z_J\}$.

Let α_J denote the $1 - 1/J$ -quantile: $\alpha_J = \Phi^{-1}(1 - 1/J)$. Using the normal tail expansion (Mills' ratio):

$$1 - \Phi(x) = \frac{\varphi(x)}{x} (1 + o(1)),$$

we have:

$$\alpha_J^2 = 2 \log J - \log 2\pi - 2 \log \alpha_J + o(1). \quad (\text{B.2})$$

For the equation to hold, we must have α_J^2 and $\log J$ having the same scale. Indeed, we have $\alpha_J^2/(2 \log J) \xrightarrow{J \rightarrow \infty} 1$, which we will verify later. Thus,

$$\log \alpha_J = \frac{1}{2} (\log 2 + \log \log J) + o(1).$$

Plugging back to Eq. B.2, we have

$$\alpha_J^2 = 2 \log J - \log 4\pi - \log \log J + o(1).$$

Taking squares via $\sqrt{A+B} = \sqrt{A}(1 + B/(2A) + o(B/A))$, we have:

$$\alpha_J = \sqrt{2 \log J} - \frac{\log 4\pi + \log \log J}{2\sqrt{2 \log J}} + o\left(\frac{1}{\sqrt{\log J}}\right).$$

Let $\beta_J = 1/\alpha_J = 1/\sqrt{2 \log J} + o(1/\sqrt{\log J})$.

Before we proceed to the analysis of the convergence of $Z_{(J)}$, we verify that $\alpha_J^2/(2 \log J) \xrightarrow{J \rightarrow \infty} 1$. Apply the Mills' ratio at $\sqrt{2 \log J}$. Then, we have

$$\begin{aligned} 1 - \Phi(\sqrt{2 \log J}) &\leq \frac{\varphi(\sqrt{2 \log J})}{\sqrt{2 \log J}} \\ &= \frac{1}{\sqrt{4\pi \log J}} \frac{1}{J} \leq \frac{1}{J}, \end{aligned}$$

where the last inequality holds for $J \geq 3$. Because $1 - \Phi(\cdot)$ is a decreasing function, $\alpha_J \leq \sqrt{2 \log J}$ for large J . That is, α_J is a slower-growing function than $\sqrt{2 \log J}$. Thus, $\log \alpha_J = O(\log \log J)$.

Plugging back to Eq. B.2, we have:

$$\alpha_J^2 = 2 \log J + \log 2\pi + O(\log \log J) \Rightarrow \frac{\alpha_J^2}{2 \log J} = o(1)$$

Next, we show that using α_J and β_J as normalizing constants, $M_J := \beta_J^{-1}(Z_{(J)} - \alpha_J)$ converges to a Gumbel distribution. To study the behavior of $\Pr(M_J \leq x) = \Phi(\alpha_J + \beta_J x)^J$, we look at the following ratio:

$$\frac{1 - \Phi(\alpha_J + \beta_J x)}{1 - \Phi(\alpha_J)} = \frac{\varphi(\alpha_J + \beta_J x)}{\varphi(\alpha_J)} \frac{T(\alpha_J + \beta_J x)}{T(\alpha_J)} = \exp\left(-x - (b_J x)^2/2\right) \frac{T(\alpha_J + \beta_J x)}{T(\alpha_J)},$$

where $T(t) := [1 - \Phi(t)]/\varphi(t)$ is the Mills' ratio. Because $\beta_J = o(1)$, we have:

$$\frac{1 - \Phi(\alpha_J + \beta_J x)}{1 - \Phi(\alpha_J)} \rightarrow \exp(-x).$$

Finally, because $J[1 - \Phi(\alpha_J)] = 1$, we have:

$$\Pr(M_J \leq x) = \Phi(\alpha_J + \beta_J x)^J = \exp(\exp(-x)).$$

Because M_J is uniformly integrable, we have:

$$\begin{aligned} \mathbb{E}[Z_{(J)}] &= \alpha_J + \gamma \beta_J \\ &= \sqrt{2 \log J} - \frac{\log 4\pi + \log \log J - 2\gamma}{2\sqrt{2 \log J}} + o\left(\frac{1}{\sqrt{\log J}}\right) \\ &= \sqrt{2 \log J} + o(1). \end{aligned}$$

Q.E.D.

Proof of Lemma 4 Define the max functional $f : \mathbb{R}^J \mapsto \mathbb{R}$ as:

$$f(x) = \max_{j \in \{1, \dots, J\}} x_j.$$

For each x , let $j^*(x)$ denote the optimizer of $\max_{j \in \mathcal{J}} x_j$ with tie-breaking rule $j^*(x) = \min \arg \max_{j \in \mathcal{J}} x_j$. The realized winner's curse is hence a mapping $wc : \mathbb{R}^J \mapsto \mathbb{R}$ defined as:

$$wc(x; \tau) = x_{j^*(x)} - \tau_{j^*(x)}.$$

For a vector of sample means, $wc(\hat{\tau}, \tau) = \hat{\tau}_{j^*} - \tau_{j^*}$ is a realized draw of the winner's curse.

Notice that wc and f share the same Hadamard directional derivative at τ , because $f(x) = wc(x) + \tau_{j^*(x)}$ and subtracting a constant does not change directional derivatives. By Rockafellar and Wets (1998), the max functional f is Hadamard directionally differentiable on $\mathbb{R}^J$. In addition,

at any τ with the maximizer set $\mathcal{J}^* = \arg \max_{j \in \{1, \dots, J\}} \tau_j$, its directional derivative in direction $h \in \mathbb{R}^J$ is $f'_\tau(h) = \sup_{j \in \mathcal{J}^*} h_j$. Therefore,

$$wc'_\tau(h) = f'_\tau(h) = \sup_{j \in \mathcal{J}^*} h_j.$$

Whenever there is more than one best action, i.e., $|\mathcal{J}^*| \geq 2$, $wc'_\tau(h)$ is nonlinear.

Finally, the (population) winner’s curse bias is $WC = \mathbb{E}_{\mathcal{D}}[wc(\hat{\tau}, \tau)]$. The expectation is taken w.r.t. the sampling distribution of the training data. Because the expectation operator is linear and continuous, so composing wc with $\mathbb{E}[\cdot]$ does not restore linearity of the derivative. Consequently, WC inherits the same lack of full Hadamard differentiability.

Q.E.D.

C. More Details on Different Correction Methods

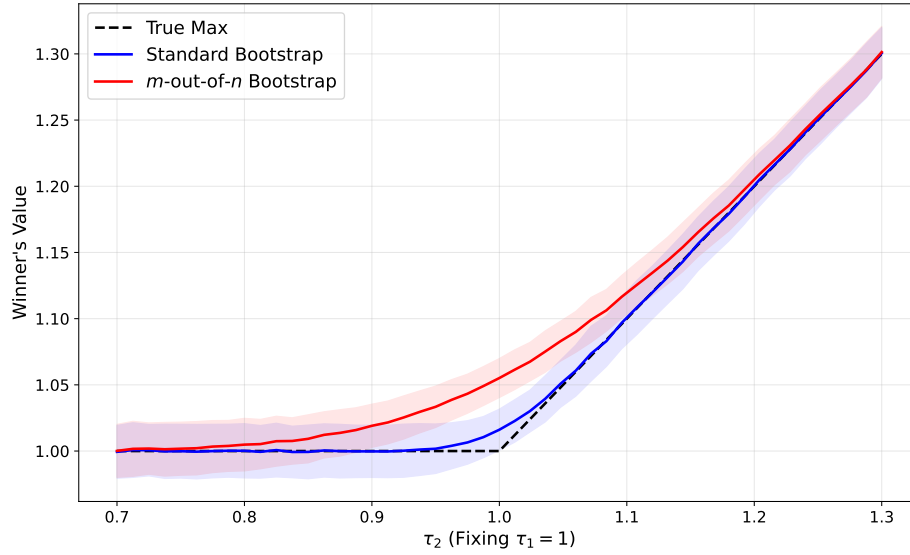
C.1. m -out-of- n Bootstrap

We use a simulation experiment to validate the intuition presented in Section 3.1. This experiment demonstrates how adjusting the resampling scale enables the m -out-of- n bootstrap to introduce sampling variability to smooth over the non-differentiable “kink” of the winner selection rule, thereby recovering valid inference.

We consider the A/B testing application in Section 4. There are two actions a_1 and a_2 , and the objective is to select the option with the higher estimated effect. Let $\hat{\tau}_j$ denote the sample average estimator for action a_j ’s true effect τ_j . The value function of interest is the expected value of the winner, $\mathbb{E}[\max\{\hat{\tau}_1, \hat{\tau}_2\}]$. Due to the maximization operator, this function is non-differentiable at the point where $\tau_1 = \tau_2$. To visualize the behavior of the standard bootstrap and the m -out-of- n bootstrap around this discontinuity, we fix the true mean of action a_1 at $\tau_1 = 1$ and vary the true mean of action a_2 across a grid from 0.7 to 1.3. Consequently, the non-differentiable “kink” occurs at $\tau_2 = 1$.

For every point on the grid, each corresponding to a unique data generation process, we draw 1,000 realizations of independent datasets. Each dataset consists of $N = 2,500$ observations per arm drawn from a normal distribution with unit variance ($\sigma = 1$). For each dataset, we calculate the expected value of the winner using the two bootstrapping methods. The standard bootstrap resamples N observations. The variance of this estimator reflects the original scale N^{-1} . The m -out-of- n bootstrap resamples $m < N$ observations. For clearer visualization, we use a more drastic smoothing factor of $m = \text{int}(N^{0.6})$.²¹ The resulting variance of this estimator increases to $m^{-1} = N^{-0.6}$.

²¹ This is only for visualization, and we use $m = \text{int}(N^{0.95})$ throughout the main analysis.

Figure C.1 Estimating the Value Function $\max\{\hat{\tau}_1, \hat{\tau}_2\}$ via Standard Bootstrap and the m -out-of- n Bootstrap

Note. Simulation parameters: $\tau_1 = 1$ and τ_2 varies from 0.7 to 1.3; Response volatility $\sigma = 1$; Sample size $N = 2,500$ per treatment. The black dashed line represents the true maximum. The blue solid line is the estimated maximum from the standard bootstrap, and the red solid line is the estimated maximum from the m -out-of- n bootstrap. Shaded regions represent ± 1 standard deviation.

Figure C.1 plots the results. We observe that the standard bootstrap (blue curve) well preserves the shape of the value function. Because the standard bootstrap treats the original dataset as the true population, even if an arm wins by chance in the original data, the standard bootstrap reproduces its winning in the simulated world with high confidence. Specifically, at the tie ($\tau_1 = \tau_2 = 1$), the standard bootstrap chooses the same winner as the original dataset 75.3% of the time. This high agreement rate indicates that the estimator rigidly commits to the empirical winner, failing to account for how “luck” plays a role in the selection at the discontinuity. In contrast, the m -out-of- n bootstrap simulates a world with less data. By adjusting the resampling scale, it introduces higher variance, preventing the estimator from committing strictly to the empirical winner. At the tie, the m -out-of- n bootstrap picks the same winner as the original dataset only 56.74% of the time. That is, about half of the bootstrap draws sits to the left of the kink and the other half sits to the right, mimicking a near-random choice. Averaging across these oscillating draws transforms the sharp kink into a curved bend (red curve). This recovered differentiability allows the estimator to correctly model the uncertainty of the winner and recover a consistent estimate of the winner’s curse bias.

The choice of the resampling scale m governs a fundamental trade-off between two competing sources of bias. When the subsample size m approaches the full sample size N (i.e., the exponent $power \rightarrow 1$), the m -out-of- n falls back to the standard bootstrap and restores the discontinuity bias.

Conversely, when m is too small, the procedure introduces excessive sampling variability that was not present in the original dataset, leading to over-correction and high estimation bias. Therefore, m must be small enough to smooth the discontinuity yet large enough to maintain estimation accuracy (Bickel et al. 1997, Bickel and Sakov 2008).

To demonstrate this empirically, we simulate 1,000 datasets for the A/B testing example where the true effects are tied ($\tau_1 = \tau_2 = 1$). We apply the bootstrap correction using various subsample size $m = \text{int}(N^{\text{power}})$, where power ranges from 0.75 to 1.0. The left panel in Figure C.2 shows that the estimation bias for the winner’s curse follows a U-shaped curve. When power is large, the discontinuity bias dominates (under-correction). When power is small, the noise-introduced bias dominates (over-correction). This mechanism is further elucidated by the agreement rate: the percentage of draws that the bootstrap winner matches the empirical winner, as shown in the right panel in Figure C.2. As power approaches one, the m -out-of- n has the same agreement rate as the standard bootstrap. While as m decreases to a small value, the agreement rate drops, effectively turning the winner selection in the bootstrap world into a coin toss at the limit and increasing the estimation bias drastically. Consequently, m must be set moderately. Because the range from $\text{int}(N^{0.9})$ to $\text{int}(N^{0.95})$ provides a reasonable balance between the two biases, we use the conservative choice of $m = \text{int}(N^{0.95})$ under different applications throughout the paper.

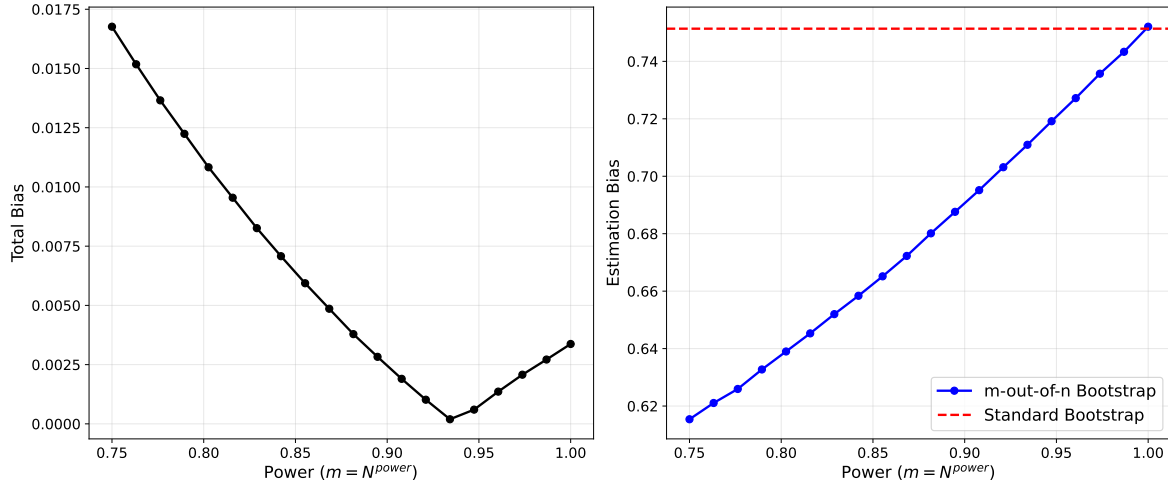
It is worth noting that for optimization landscapes more complicated than simply picking the highest score, the nature of the discontinuity may be more sophisticated. In such cases, stronger smoothing (a smaller m) might be required (Chakraborty et al. 2013). However, because there is currently no consensus on how to set the smoothing factor robustly for general applications, we adhere to our conservative choice throughout our paper and leave the adaptive design of subsampling sizes for future research.

C.2. Sample Splitting

In this section, we show that when using half of the dataset, how much the sample splitting decision loses compared to using the entire dataset. Consider the same setting as Section 4, where we have two treatments a_1, a_2 with treatment effects τ_1, τ_2 . We assume without loss of generality that $\tau_2 > \tau_1$ and define $\Delta\tau = \tau_2 - \tau_1$. Given a dataset with N observations for each treatment, we construct independent normal estimators $\tau_1 \sim \mathcal{N}(\tau_1, \sigma^2/N)$ and $\tau_2 \sim \mathcal{N}(\tau_2, \sigma^2/N)$ for the two treatments, respectively.

The actual value of the winning treatment picked from the entire dataset is:

$$\begin{aligned} \text{Val}(N) &= \mathbb{E}[\tau_1 \mathbb{1}\{\hat{\tau}_1 > \hat{\tau}_2\}] + \mathbb{E}[\tau_2 \mathbb{1}\{\hat{\tau}_1 \leq \hat{\tau}_2\}] \\ &= \tau_1 + (\tau_2 - \tau_1) \mathbb{E}[\mathbb{1}\{\hat{\tau}_2 \geq \hat{\tau}_1\}] \\ &= \tau_1 + \Delta\tau \Phi\left(\sqrt{\frac{N}{2}} \frac{\Delta\tau}{\sigma}\right). \end{aligned}$$

Figure C.2 Impact of the Resampling Scale m 

Note. Simulation parameters: $\tau_1 = \tau_2 = 1$; Response volatility $\sigma = 1$; Sample size $N = 2,500$ per treatment. $m = \text{int}(N^{\text{power}})$. The left panel plots the total bias as a function of the smoothing parameter *power*. The right panel visualizes how estimation bias, i.e., probability of bootstrap winner being the same as the empirical winner, changes with *power*.

where Φ is the CDF of the standard normal distribution. When using half of the dataset to pick the better treatment, the optimality gap, as a percentage of the actual lift $\Delta\tau$, is:

$$\begin{aligned} SSGap &:= \Delta\tau^{-1} \left[Val(N) - Val\left(\frac{N}{2}\right) \right] \\ &= \Phi\left(\sqrt{\frac{N}{2}} \frac{\Delta\tau}{\sigma}\right) - \Phi\left(\sqrt{\frac{N}{4}} \frac{\Delta\tau}{\sigma}\right). \end{aligned}$$

Next, we examine how the optimality gap from sample splitting varies with the signal-to-noise ratio. Let $\rho = \Delta\tau/\sigma$ denote the signal-to-noise ratio. The derivative of the gap with respect to ρ is given by:

$$\frac{d SSGap}{d\rho} = \sqrt{\frac{N}{2}} \left[\varphi\left(\sqrt{\frac{N}{2}}\rho\right) - \frac{1}{\sqrt{2}} \varphi\left(\sqrt{\frac{N}{4}}\rho\right) \right].$$

Solving the first order condition shows that $SSGap$ initially increases with ρ , reaches a maximum at

$$\rho_{max} = 2\sqrt{\frac{\ln 2}{N}},$$

and then declines toward zero as the signal-to-noise ratio strengthens. When we have $N = 2,500$ observations per treatment, the $SSGap$ attains its maximum at $\rho_{max} = 3.33\%$. Substituting this value back into the expression for $SSGap$ reveals that halving the dataset leads to an optimality gap of at most $\Phi(\sqrt{2\ln 2}) - \Phi(\ln 2) = 8.3\%$ of the true lift $\Delta\tau$.

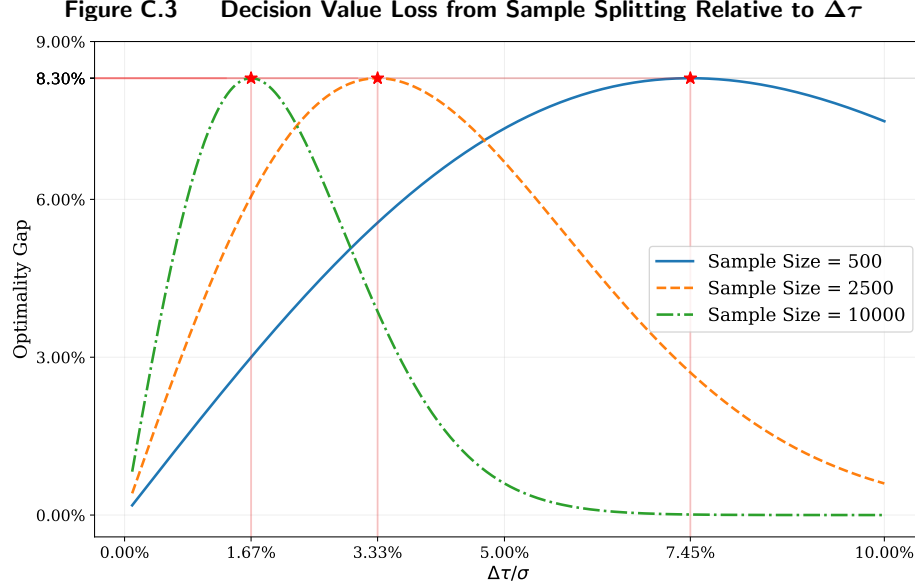


Figure C.3 illustrates how the optimality gap from sample splitting evolves as a function of the signal-to-noise ratio across three different sample sizes. When the signal-to-noise ratio is small, doubling the dataset will not significantly improve decision value because the signal is buried in too much noise. On the contrary, when the signal-to-noise ratio is huge, doubling the dataset will not improve the decision value either, because using a small amount of data is sufficient to identify the better treatment. The value of additional data is the largest when the signal-to-noise ratio is moderate. In this case, doubling the sample size makes a crucial difference in overcoming the noise and identifying the signal.

Comparing across sample sizes, we find that the optimality gap persists over a broader range of signal-to-noise ratios when the sample size is small (e.g., $N = 500$ per treatment). In contrast, when the sample size is large (e.g., $N = 10,000$ per treatment), the gap is smaller across most of the spectrum except the peak. Interestingly, while the location of the peak gap shifts with the sample size, its magnitude remains constant: the gap reaches its maximum of 8.3% at $\rho = 7.45\%$ when $N = 500$ per arm and at $\rho = 1.67\%$ when $N = 10,000$ per arm.

C.3. Bayesian Methods

In this section, we show that if a firm chooses to apply Bayesian methods to correct for the winner's curse, the empirical Bayes approach is generally preferable to using a fixed prior, as the latter can in some cases exacerbate the winner's curse. To illustrate this, consider the A/B testing scenario in Section 4, where there are two actions $\mathcal{A} = \{a_1, a_2\}$ with true rewards τ_1 and τ_2 . A canonical Bayesian model with a fixed prior assumes that τ_j is drawn from a common normal prior $\mathcal{N}(\mu, s^2)$

Table C.1 Two Treatments: Bayesian Estimation Under Positive vs. Negative Treatment Effects

(τ_1, τ_2)	(1, 1.01)	(-1.01, -1)
No Correction	113.85% (171.14%)	113.85% (171.14%)
Bayesian	109.78% (171.07%)	117.81% (171.08%)
Empirical Bayes	45.25% (161.82%)	45.25% (161.82%)
Standard Bootstrap	37.49% (190.41%)	37.49% (190.41%)
m-out-of-n Bootstrap	11.65% (191.38%)	11.65% (191.38%)

Simulation parameters: Response noise level $\sigma = 1$; Sample size $N = 2,500$; Results aggregate from 1,000 repetitions. $m = \text{int}(N^{0.95})$ for m -out-of- n bootstrap. The prior distribution for the Bayesian method is $\mathcal{N}(0, 1)$;

(Smith and Winkler 2006, Kessler 2024), and that we can construct MLE estimators $\hat{\tau}_j$ following $\mathcal{N}(\tau_j, \sigma_s^2)$. Under this specification, the posterior mean for each option is:

$$\hat{\tau}_j^{post} = \frac{s^2}{s^2 + \sigma_s^2} \hat{\tau}_j + \frac{\sigma_s^2}{s^2 + \sigma_s^2} \mu, \forall j \in \{1, 2\}.$$

This expression shows that the posterior mean shrinks each MLE estimate toward the prior mean μ . When μ is below the true rewards τ_j , this shrinkage moves estimates in the correct direction, reducing overestimation and thereby mitigating the winner’s curse.

However, when the assumed prior mean μ is above the true rewards, the “shrinkage” moves estimates in the wrong direction, increasing the winner’s curse rather than correcting it. To illustrate, we consider two data generating processes with the same signal-to-noise ratio at 1%, but one with positive true rewards and one with negative true rewards (Table C.1). Using a mean-zero normal prior, such as in Smith and Winkler (2006), the posterior shrinks toward zero in both cases. When the true rewards are positive, this shrinkage reduces the winner’s curse modestly. But when the true rewards are negative, the same “shrinkage” toward zero makes the estimates even more optimistic. As a result, the winner’s curse increases from 113.85% without correction to 117.81% when using a fixed-prior Bayesian.

C.4. Selective Inference

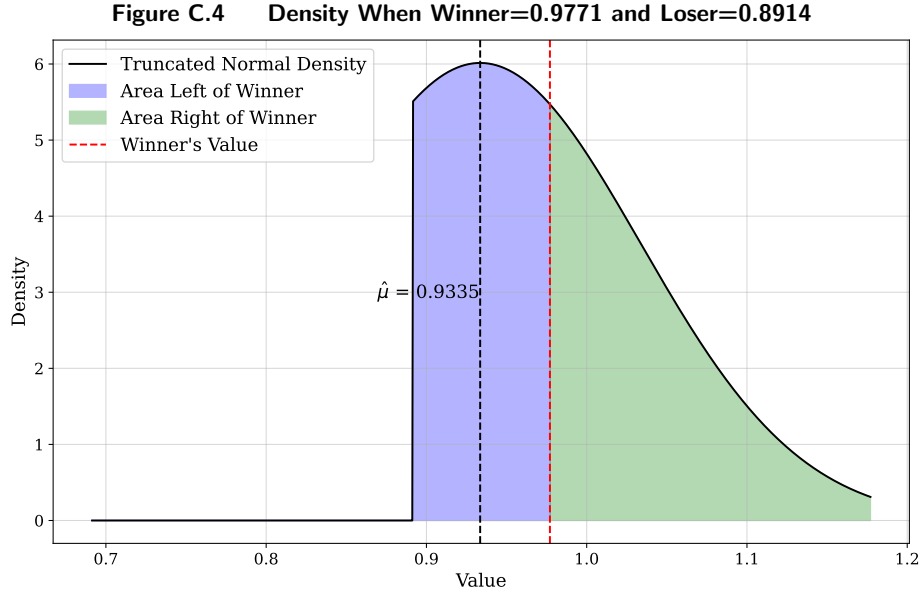
First, we summarize the algorithm for the conditional selective inference algorithm proposed by (Andrews et al. 2024) in Algorithm C.1.

Next, we provide a graphical explanation of how the conditional selective inference estimator from (Andrews et al. 2021) works. Consider a simple scenario with two candidate treatments a_1

Algorithm C.1 Conditional Selective Inference Correction (Andrews et al. 2024)

Require: Dataset $\mathcal{D}$ with N observations; Number of bootstraps B .

- 1: Calculate the estimates $\hat{\theta}_{1:J}$ and their sampling variance $\hat{\sigma}_j^2$ from $\mathcal{D}$
 - 2: Pick the largest estimate $\mathbf{T}^*(\hat{\theta}_{1:J}) = \operatorname{argmax}_{j \in \{1, \dots, J\}} \hat{\theta}_{1:J}$
 - 3: Find the second largest estimate $L = \max_{j \neq j^*} \hat{\theta}_{1:J}$
 - 4: Solve $F_{TN}(\hat{\theta}_{j^*}; \mu, \sigma_{j^*}, L) = 0.5$ for the root $\hat{\mu}$
 - 5: **return** $\hat{\mu}$
-

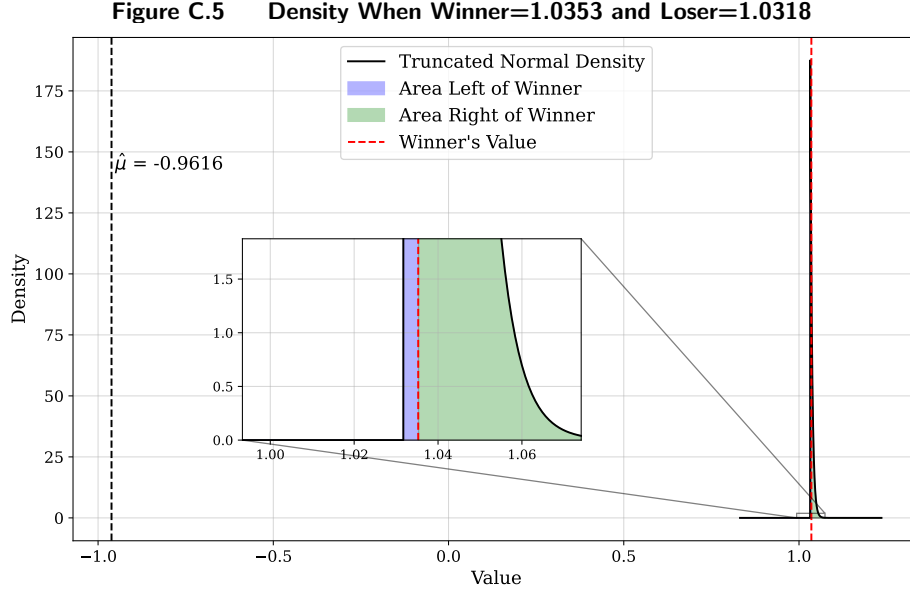


and a_2 , whose true effects are $\tau_1 = 1$ and $\tau_2 = 1.1$, respectively. We draw 100 observations for each arm, each following a normal distribution with the treatment effect as the mean and a known variance of one. That is, $Y_{ij} \sim \mathcal{N}(\tau_j, 1)$ for observations $i \in \{1, \dots, 100\}$ and treatments $j \in \{1, 2\}$. We want to pick the treatment with a higher estimated effect $\hat{\tau}_j$, which we call the winner, and we will refer to the treatment with the smaller effect as the loser.

In one of the realizations, the estimated treatment effects are $\hat{\tau}_1 = 0.8914$ and $\hat{\tau}_2 = 0.9771$, with a difference of 0.0875. The way Andrews et al. (2024)'s formula

$$F_{TN}(\hat{\theta}_{j^*}; \mu, \sigma_{j^*}, L) = 0.5,$$

works for finding a median-unbiased estimator is to move the location μ of the truncated normal distribution until the left and right (w.r.t. the winner's value) areas under the density curve are the same (equal to 0.5), as shown in Figure C.4. In this case, the estimator works nicely and finds that the median-unbiased estimate for the winner is 0.9335.



However, the estimator becomes unstable when the two estimated treatment effects are close, as seen in all the experiments in the main article. To see this analytically, we can write the estimated effects as $\hat{\tau}_2 = \hat{\tau}_1 + \delta$ and assume without loss of generality that $\hat{\tau}_2 \geq \hat{\tau}_1$, i.e., $\delta \geq 0$. As δ approaches zero from above, the CDF of the truncated normal (left-hand side) will go to zero:

$$\lim_{\delta \downarrow 0} F_{TN}(\hat{\tau}_2) = \frac{\Phi\left(\frac{\hat{\tau}_1 + \delta - \mu}{\sigma_s}\right) - \Phi\left(\frac{\hat{\tau}_1 - \mu}{\sigma_s}\right)}{1 - \Phi\left(\frac{\hat{\tau}_1 - \mu}{\sigma_s}\right)} = 0,$$

where σ_s is the sampling volatility of the estimated effects. As a result, when the two estimated effects are sufficiently close, the equation will become infeasible to solve. To see this graphically, we find in another realization where the estimated effects are $\hat{\tau}_1 = 1.0318$ and $\hat{\tau}_2 = 1.0353$, with a difference of 0.0035. As shown in Figure C.5, for the left and right areas under the density curve to be equal (to 0.5), because the loser's value is so close to the winner's value, the location of the truncated normal distribution must move to the far left. As a result, the resulting estimate for the winner after correction is -0.9616, which significantly over-corrects.

To see the problem from another angle, we plot the root-finding equation $F_{TN}(\hat{\theta}_{j*}; \mu, \sigma_{j*}, L) - 0.5$ under different levels of realized differences between the two treatments. As shown in Figure C.6, when the realized difference is large, the estimator performs reasonably well. However, as the difference shrinks, the solution to the equation quickly deteriorates. Finally, we plot the corrected winner's value as a function of the realized difference in Figure C.7. When the realized difference falls below 0.02, this selective inference estimator will drop to the zone of over-correction.

Figure C.6 Root-Finding For $F_{TN}(\hat{\theta}_{j^*}; \mu, \sigma_{j^*}, L) = 0.5$ Under Different Realized Differences Between Two Estimated Treatment Effects

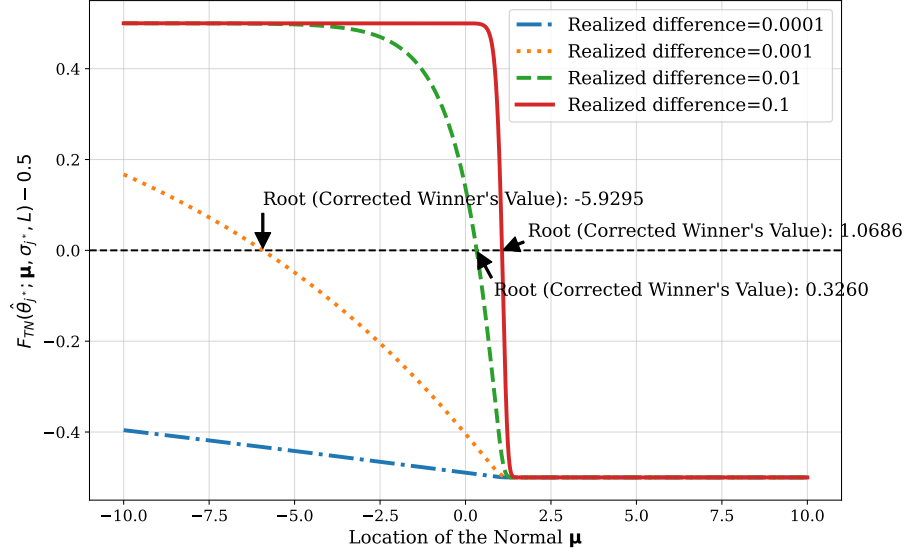
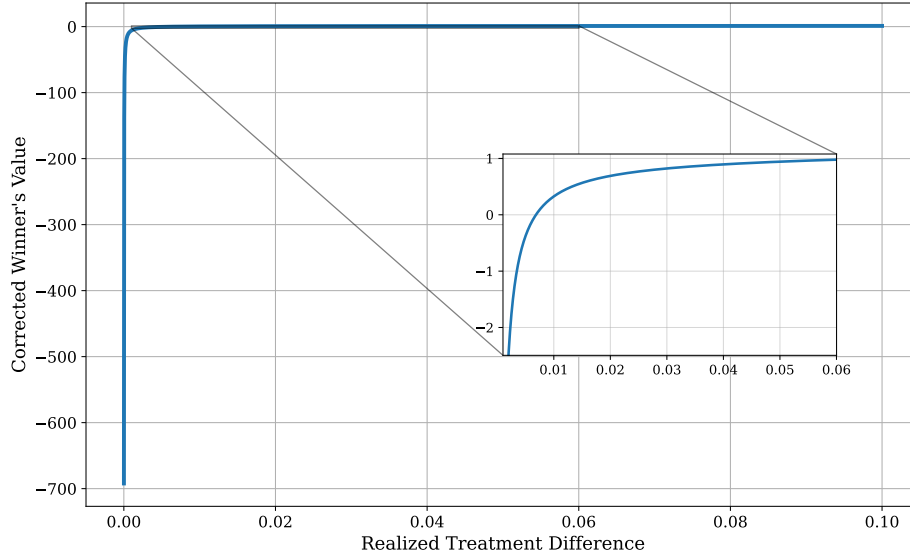


Figure C.7 Corrected Winner's Curse vs. Realized Differences Between Two Estimated Treatment Effects



D. Supplementary Results

D.1. More Estimators in A/B Testing

In addition to the estimators presented in Table 2 in the main article, we further compare the performance of four more estimators. The first one is a plugin estimator. As discussed in Lemma 2, when we are choosing from two candidate treatments, there is an analytical formulation for the winner's curse:

$$WC = \sigma \sqrt{\frac{2}{N}} \varphi\left(\sqrt{\frac{N}{2}} \cdot \frac{\Delta\tau}{\sigma}\right)$$

Table D.2 **A/B Testing: Average Winner's Curse $\bar{W}C/\Delta\tau$ Under Different $\Delta\tau/\sigma$**

$\Delta\tau/\sigma$	0.5%	1%	2%
No Correction	238.90% (333.83%)	113.85% (171.14%)	47.03% (90.94%)
Conditional	-280.76%	-191.40%	-81.18%
Selective Inference	(1673.52%)	(1600.16%)	(489.22%)
Plugin Correction	-47.07% (422.37%)	-20.39% (214.75%)	-16.12% (114.99%)
Sample Splitting	0.68%	-0.58%	3.48%
(90-10 Split)	(1272.34%)	(630.36%)	(313.14%)
10-Fold CV	11.55% (307.25%)	-2.29% (159.40%)	-14.97% (87.09%)
Jackknife	67.56% (1265.21%)	4.86% (672.30%)	3.77% (318.60%)

Simulation parameters: $\tau_1 = 1$, $\tau_2 = \tau_1 + \Delta\tau$; Response noise level $\sigma = 1$; Sample size $N = 2,500$, Results aggregate from 1,000 repetitions. $m = \text{int}(N^{0.95})$ for m -out-of- n bootstrap. Numbers in the parentheses are **standard deviations** of the winner's curse.

A natural estimator is to replace the unknown $\Delta\tau$ and σ by their empirical estimates. Subtracting this estimated winner's curse from the decision value estimate yields the plugin estimator. As shown in Table D.2, the plugin estimator reduces the winner's curse to 71.86% when the signal-to-noise ratio is 0.5%, roughly the same magnitude as the standard bootstrap.

The next group of estimators is variations of the 50/50 split sample splitting. First, we consider sample splitting with a 90/10 split. When we increase the portion of data used for picking the better treatment, the suboptimal treatment chosen by sample splitting agrees with the focal (optimal) treatment learned from the entire dataset in around 95% of the 500 repeated experiments, compared to 80% of sample splitting with a 50/50 split. However, because we are using less data for evaluation, it suffers from larger standard errors, around twice that of the 50/50 sample splitting and three times that of the no correction estimator.

Another variation is K -fold Cross-Validation (CV). Specifically, it partitions the dataset into K subsets, iteratively uses $K - 1$ subsets to select the better treatment, evaluates the selected option on the remaining subset, and finally averages the K evaluations. However, we note that because the decisions picked in different iterations are potentially different, the average of all evaluations does not have a clear interpretation of which decision we are evaluating exactly. Nevertheless, we include the results in Table D.2 for completeness. To address such a problem, one can use a Jackknife estimator, which partitions the data into N subsets and leaves only one observation for evaluation in each iteration. By doing this, the decision picked in each iteration will most likely be the same

as the focal decision learned from the entire dataset. However, because it uses one observation pair for evaluation, it suffers from large variances and is highly sensitive to outliers.

D.2. Imbalanced Sample Sizes

Throughout the main analysis, we assumed a balanced sample size across actions. When sample sizes are imbalanced, however, the action with fewer observations has a larger estimation variance. This inflated variance increases the likelihood that this action obtains an overly large estimate by chance, thereby amplifying the winner’s curse.

To illustrate this effect, reconsider the A/B testing setup with two actions a_1, a_2 . Suppose we now observe N_1 samples for a_1 and N_2 samples for a_2 , with the total number of observations held fixed at the balanced benchmark: $N_1 + N_2 = 2N$. A direct extension of the analysis in Lemma 2 shows that the winner’s curse under imbalance is:

$$WC = \sigma \sqrt{\frac{N_1 + N_2}{N_1 N_2}} \varphi \left(\frac{\Delta\tau}{\sigma} \sqrt{\frac{N_1 N_2}{N_1 + N_2}} \right).$$

This expression shows that the winner’s curse is minimized when the experiment is perfectly balanced ($N_1 = N_2$) and grows as the allocation becomes more uneven. To see this more clearly, we follow the same simulation procedure as in Section 4 but this time vary the sample sizes across treatments. Table D.3 shows that as soon as the sample sizes deviate from equality (e.g., $N_1 = 4,500$ and $N_2 = 500$ instead of $N_1 = N_2 = 2,500$), the winner’s curse becomes substantially larger, increasing from 238.90% to 388.22%.

When sample sizes differ across treatments, the bootstrap correction is implemented in a stratified manner: we resample within each treatment group separately. For example, when $(N_1, N_2) = (4,500, 500)$, we draw 4,500 observations with replacement from the a_1 group and 500 observations from the a_2 group. The m -out-of- n bootstrap is applied in the way. In the same example, we resample $\text{int}(4,500^{0.95})$ observations from action a_1 and $\text{int}(500^{0.95})$ observations from action a_2 . Table D.4 reports the resulting winner’s curse under this imbalanced design for the bootstrap estimator as well as the three alternative correction methods discussed in the main analysis. The conclusion remains unchanged: the (stratified) bootstrap-based methods continue to deliver strong performance and outperform the other correction approaches even when the experiment is imbalanced.

D.3. Personalized Targeting: Impact of Different Hyperparameters

The winner’s curse when using the causal forest (max depth five) to estimate the heterogeneous treatment effects under different signal-to-noise ratios is summarized in Table D.5.

The functional form of how customer characteristics affect their heterogeneous treatment effects can affect the winner’s curse. To see this, we consider four different forms of the g function in

Table D.3 A/B Testing: Imbalanced Sample Sizes

(N_1, N_2)	$\Delta\tau/\sigma = 0.50\%$	$\Delta\tau/\sigma = 1.00\%$	$\Delta\tau/\sigma = 2.00\%$
(500, 4500)	383.61% (547.19%)	188.97% (275.18%)	87.41% (140.10%)
(1000, 4000)	301.04% (411.28%)	146.35% (207.14%)	66.75% (107.22%)
(1500, 3500)	266.57% (358.91%)	128.71% (182.89%)	56.99% (95.06%)
(2000, 3000)	249.38% (339.63%)	120.80% (171.73%)	51.28% (90.12%)
(2500, 2500)	238.90% (333.83%)	113.85% (171.14%)	47.03% (90.94%)
(3000, 2000)	237.12% (334.86%)	114.73% (170.11%)	49.91% (90.25%)
(3500, 1500)	254.74% (353.36%)	123.54% (179.20%)	52.59% (96.46%)
(4000, 1000)	286.34% (400.69%)	139.73% (203.56%)	62.46% (108.42%)
(4500, 500)	388.22% (539.75%)	191.24% (272.42%)	89.99% (141.28%)

Simulation parameters: $\tau_1 = 1$, $\tau_2 = \tau_1 + \Delta\tau$; Response noise level $\sigma = 1$; Sample sizes per treatment are (N_1, N_2) with $N_1 + N_2 = 5,000$; Results aggregate from 1,000 repetitions; $m = \text{int}(N^{0.95})$ for m -out-of- n bootstrap. Numbers in the parentheses are **standard deviations** of the winner's curse.

Table D.4 A/B Testing: Average Winner's Curse $\bar{WC}/\Delta\tau$ With an Imbalanced Dataset

$\Delta\tau/\sigma$	0.5%	1.0%	2.0%
No Correction	388.22% (539.75%)	191.24% (272.42%)	89.99% (141.28%)
Standard Bootstrap	126.72% (608.48%)	61.50% (310.60%)	27.12% (163.47%)
m-out-of-n Bootstrap	56.23% (612.01%)	26.36% (312.70%)	9.74% (164.98%)
Sample Splitting	37.88% (908.06%)	13.26% (465.18%)	7.56% (243.80%)
Empirical Bayes	188.74% (444.67%)	85.83% (224.48%)	31.35% (118.22%)
Hybrid	-114.20%	-60.06%	-35.49%
Selective Inference	(1014.21%)	(525.19%)	(280.28%)

Simulation parameters: $\tau_1 = 1$, $\tau_2 = \tau_1 + \Delta\tau$; Response noise level $\sigma = 1$; $N_1 = 4,500$ for action a_1 and $N_2 = 500$ for action a_2 ; Results aggregate from 1,000 repetitions; $m = \text{int}(N^{0.95})$ for m -out-of- n bootstrap. Numbers in the parentheses are **standard deviations** of the winner's curse.

Table D.5 Personalized Targeting: Average Winner’s Curse $\bar{W}C/\Delta\tau$ Under Different $\Delta\tau/\sigma$ When Using Causal Forests

$\Delta\tau/\sigma$	0.5%	1%	2%
No Correction	690.27% (511.22%)	342.41% (255.87%)	165.76% (128.36%)
Standard Bootstrap	186.66% (562.33%)	91.43% (281.33%)	41.83% (140.95%)
m-out-of-n Bootstrap	23.28% (565.06%)	9.70% (282.49%)	1.41% (141.35%)
Sample Splitting	-9.80% (773.37%)	-6.35% (387.22%)	-4.26% (194.46%)
Empirical Bayes	246.94% (517.30%)	120.93% (259.21%)	55.18% (130.02%)
Hybrid	-2855.93%	-1425.85%	-705.55%
Selective Inference	(1001.69%)	(500.32%)	(248.95%)

Simulation parameters: $\tau_1 = 1$, $\tau_2 = \tau_1 + \Delta\tau$; $g = x^2 + x$; Response noise level $\sigma = 1$; Sample size is $N = 2,500$ per treatment; Results aggregate from 1,000 repetitions; Causal forest has max depth of 5; $m = \text{int}(N^{0.95})$ for m -out-of- n bootstrap. Numbers in the parentheses are **standard deviations** of the winner’s curse.

Table D.6. Note that the impact of the functional form of heterogeneity on the winner’s curse is confounded with the distribution of customer characteristics, which is a standard normal in this case. As a result, when we use the cubic function $g(x) = x^3$, because many customers will have close to zero characteristics, they will have very close treatment effects. Consequently, it is difficult to identify the better treatment for those customers, and the winner’s curse is large.

D.4. Median Winner’s Curse

To better understand the shape of the winner’s curse distribution, we provide the median of the realized winner’s curse across the 1,000 simulation draws for all result tables in this paper.

Table D.6 Personalized Targeting: Average Winner's Curse
 $\bar{WC}/\Delta\tau$ Differs with the Functional Form of Heterogeneity

	x	$x^2 + x$	$ x $
No Correction	373.20% (248.78%)	342.41% (255.87%)	340.76% (250.50%)
Standard Bootstrap	67.78% (270.91%)	91.43% (281.33%)	75.04% (272.89%)
m-out-of-n Bootstrap	-5.36% (275.65%)	9.70% (282.49%)	3.32% (276.95%)
Sample Splitting	12.52% (376.06%)	-6.35% (387.22%)	9.23% (377.70%)
Empirical Bayes	360.03% (249.88%)	120.93% (259.21%)	118.82% (251.97%)
Hybrid	-1106.71%	-1425.85%	-930.50%
Selective Inference	(390.75%)	(500.32%)	(394.94%)

Simulation parameters: $(\tau_1, \tau_2) = (1, 1.01)$; Response volatility $\sigma = 1$; Training set has $N = 2,500$ observations per arm; Out-of-sample customers $K = 10,000$; Results aggregate from 1,000 draws; Causal forest estimator has max depth of 5; $m = \text{int}(N^{0.95})$ for m -out-of- n bootstrap. Numbers in the parentheses are **standard deviations** of the winner's curse.

Table D.7 A/B Testing: Median Winner's Curse
 $\text{med}(WC)/\Delta\tau$ Under Different $\Delta\tau/\sigma$

$\Delta\tau/\sigma$	0.5%	1%	2%
No Correction	213.66%	103.42%	44.49%
Standard Bootstrap	49.85%	23.53%	7.92%
m-out-of-n Bootstrap	-1.76%	-5.66%	-5.44%
Sample Splitting	17.29%	7.80%	3.00%
Empirical Bayes	90.75%	40.79%	5.76%
Hybrid	6.52%	6.66%	2.74%
Selective Inference			

Simulation parameters: $\tau_1 = 1$, $\tau_2 = \tau_1 + \Delta\tau$; Response noise level $\sigma = 1$; Sample size $N = 2,500$; Results aggregate from 1,000 repetitions. $m = \text{int}(N^{0.95})$ for m -out-of- n bootstrap.

Table D.8 A/B Testing: Median Winner's Curse $\text{med}(\bar{W}C)/\Delta\tau$ Under Difference Response Distributions with $\Delta\tau/\sigma = 1\%$

	Normal(0, 1)	Uniform ($-\sqrt{3}, \sqrt{3}$)	Laplace (0, $1/\sqrt{2}$)	Logistic (0, $\sqrt{3}/\pi$)
No Correction	103.42%	110.06%	96.52%	101.02%
Standard Bootstrap	23.53%	31.51%	18.39%	26.18%
m-out-of-n Bootstrap	-5.66%	3.12%	-8.89%	0.23%
Sample Splitting	7.80%	11.99%	-13.80%	-3.39%
Empirical Bayes	40.79%	35.12%	28.48%	32.18%
Hybrid				
Selective Inference	6.66%	21.54%	1.59%	10.53%

All distributions have variance one. Simulation parameters: $(\tau_1, \tau_2) = (1, 1.01)$; Sample size $N = 2,500$ per treatment; Results aggregate from 1,000 repetitions. $m = \text{int}(N^{0.95})$ for m -out-of- n bootstrap.

Table D.9 Personalized Targeting: Median Winner's Curse $\text{med}(WC)/\Delta\tau$ Under Correct Model Specification vs. Causal Forest

	Correct Functional Form	Causal Forest (Max Depth = 2)	Causal Forest (Max Depth = 5)	Causal Forest (Max Depth = 10)
No Correction	85.46%	228.56%	337.13%	818.19%
Standard Bootstrap	4.42%	87.55%	84.61%	211.18%
m-out-of-n Bootstrap	-28.45%	30.93%	5.18%	110.29%
Sample Splitting	-1.48%	-30.80%	-5.92%	-9.86%
Empirical Bayes	77.70%	-235.44%	115.34%	557.27%
Hybrid				
Selective Inference	-47.74%	-1825.99%	-1435.37%	-1555.57%

Simulation parameters: $(\tau_1, \tau_2) = (1, 1.01)$; Response noise level $\sigma = 1$ so the signal-to-noise ratio $\Delta/\sigma = 1\%$; $g(x) = x^2 + x$; Sample size is $N = 2,500$ per treatment; Results aggregate from 1,000 repetitions; $m = \text{int}(N^{0.95})$ for m -out-of- n bootstrap.